

IDENTIFICATION

PRODUCT NAME: AD8E, AM8E A-D CONVERTER AND  
MULTIPLEXER DIAGNOSTIC

PRODUCT CODE: MAINDEC-08-DHADA-A-D

FORMERLY: MAINDEC-8E-D68B-D-(D)

DATE REVISED: MARCH 1972

MAINTAINER: DIAGNOSTIC GROUP

AUTHOR: P.T. COYNE

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READ THIS DOCUMENT PRIOR TO RUNNING PROGRAM:

## 1. ABSTRACT

This program performs basic tests on the Input/Output control logic and multiplexer. The analog tests are designed to provide a means of calibrating the converter and checking conversion parameters.

## 2. REQUIREMENTS

### 2.1 Equipment

PDP-8/E with 4K core, ASR33 teletype, AD8E A-D Converter, (AM8E Multiplexer optional), Adjustable High Quality Voltage Source, EDC MODEL MV105G or equivalent.

NOTE: To run MONOTONICITY TEST, a function generator capable of .1 CPS, sine or ramp output must be used.

### 2.2 Storage

Maindec resides in locations 0000-4500.

### 2.3 Preliminary Programs

All basic CPU and teletype Maindecs must have been run successfully.

NOTE: If external enable utilizing the DK8-E REAL TIME CLOCK is to be run, the Maindec for the DK8-E must be successfully run first. In addition, VC8-E Control Tests must be run prior to special LAB-E SYSTEM CHECK routine.

## 3. LOADING PROCEDURE

The binary loader is used to load the program.

## 3.1 Control Switches

- SW0 = Suppress error messages and "END LOGIC TEST" MESSAGE
- SW1 = HALT ON ERROR WITH PC displayed in AC,
- SW2 = Scope loop override to exit from loop on error and permit continuance of test. Also halts with converted word in AC for EXTERNAL ENABLE when there is no error.
- SW3 = Enables halt during calibration routine. Converted word is displayed in AC.
- SW4 = Must be set to run EXTERNAL ENABLE test.
- SW5 = Allows operator to explicitly select any one of the logic routines.

## 4. USAGE PROCEDURE

\*\*\*SEE SPECIFICATIONS FOR MAXIMUM VOLTAGE INPUTS!\*\*\*

\*\*\*INSURE THAT TELETYPE IS ON-LINE.\*\*\*

## a. Control Logic Test

1. LOAD 200.
2. Press CLEAR then CONTINUE. HALT will occur.
3. Select options from switches 0, 1, 2, 5.
4. If SW5 is present (1), select test from SW8-11.
5. Press CONTINUE.
6. After each pass (12 sec) "END OF LOGIC TEST" will be printed.

NOTE: With SW5 down and SW2 up, any error will be reported once, then program will continue to next test.

## b. IOT Scope Loop

1. LOAD 201.
2. Place low order six bits of IOT 65xx in SW6-11.
3. Press CLEAR, then CONTINUE.

NOTE: IOT may be reselected while running.

c. Display Converted Value In AC.

1. Apply voltage to A-D converter Input or to multiplexer channel Inputs.
2. LOAD 202.
3. If a HALT after conversion is desired, select SW3.
4. Select MPX channel from SW8-11; Select channel 0 if no multiplexer is available.
5. Press CLEAR, the CONTINUE; the converted value will be observed in the AC.
6. When SW3 halt select is engaged, operator may change channels. If desired, then press CONTINUE to loop. SW3 may be deselected at this time.

d. External Enable with Real Time Clock (DK8EP or DK8ES)

1. Apply voltage to A-D Converter Input or preamplifiers, if desired.
2. LOAD 203.
3. Set SW4.
4. Select switches 0 or 2 as desired.
5. Select channel with SW 8-11.
6. Press CLEAR, then CONTINUE.
7. After each pass the TTY bell will ring.

NOTE: Channel may be changed while running test.

e. Monotonicity Test

NOTE: Ramp Speed of function generator must be slower than slew rate of converter. See ENGINEERING SPECIFICATIONS. (.1 HZ is a good setting).

1. Connect function generator to CHNL 0 or to AD8E Input.

2. LOAD 204
3. Select SW0 If desired.
4. Press CLEAR, then CONTINUE.
5. Program will halt.
6. Select Stall time between tests Iterations by selecting SW0-11. The larger the number in the switch register, the greater the stall time.
7. Press CONTINUE.
8. If error occurs, program will halt with word "N" In AC. Pressing CONTINUE will display "N+1" word In AC. Pressing CONTINUE again will restart test.

f. Resolution Accuracy Test

1. Apply a known voltage to A-D converter Input.
2. LOAD 205.
3. Select SWS 0,1 If desired.
4. Select channel with SW8-11.
5. Press CLEAR, then CONTINUE.
6. If error occurs, program will typeout the two non-comparing words on TTY then continue with test.
7. If no error occurs, TTY bell will ring once then, program will recycle. One cycle being 500,000(10) conversions.

g. Successive Reads Test

1. Apply any voltage to A-D converter Inputs at preamplifier.
2. LOAD 206.
3. Select SW0 If desired.
4. Select channel from SW8-11.
5. Press CLEAR, then CONTINUE.

6. If error occurs, program will halt with first read in AC. Press CONTINUE to get second read into AC.
7. To restart, press continue.
8. If no error occurs, TTY bell will ring once, then program will recycle.

#### h. Multiplexer noise test

1. LOAD 207.
2. Select channel in SW8-11 and apply voltage to that channel.
3. Select SW0 if desired.
4. Press CLEAR, then CONTINUE.
5. If error occurs, message will be typed on TTY, then routine will recycle.

#### i. LAB8-E System Test

The system must contain a DK8-EP option and a VC8-E option with a display.

1. Apply a voltage input to the A/D or multiplexer.
2. LOAD 210.
3. Depress CLEAR, CONTINUE.
4. Program will halt.
5. Select clock frequency via SW3-5, reference LAB8-E programming card for DK8-EP clock rate. (1MHZ=6, 100KHZ=5, ..., 100HZ=2).
6. Press CONTINUE then observe printout:  
"SET SW5(AUTO=INC), NUMBER OF CHNLS IN SW8-11 OR SET SW8-11 (SINGLE CHNL)".
7. If all channels are to be displayed at the same time, set SW5, then set the number of channels contained within the system into SW8-11, I.E., IF SYSTEM CONTAINS ONE A232, SET THE SWITCH REGISTER TO 0110. IF ONLY CHNL FOUR IS TO BE OBSERVED SET THE SWITCH REGISTER TO 0004.

8. DEPRESS CONTINUE and observe the display scope. A horizontal line should be present for channel selected. By varying the input voltage the line should move up or down.  $0V \approx 1/2$ -screen,  $+V = TOP$ ,  $-V = bottom$ . A sweep of the scope is generated on each clock overflow. Thus it is a function of the clock rate set in (5).

## 5. PROGRAM DESCRIPTION

### 5.1 Control Logic Tests

Consists of 14 separate checks to assure the control logic is functioning properly.

- TST0 = Checks that A-D DONE and TIMING ERROR flags are cleared by Initialize,
- TST1 = Checks that A-D DONE flag can be set then cleared.
- TST2 = Checks that TIMING ERROR flag can be set then cleared.
- TST3 = Test for unexpected Interrupt request.
- TST4 = Tests to see if ADRB Jam transfers to AC.
- TST5 = Tests to see if ADRS Jam transfers to AC.
- TST6 = Tests to see if enable register can be loaded and read back.
- TST7 = Tests to see if A-D DONE will generate Interrupt.
- TST10 = Tests to see if TIMING ERROR will generate Interrupt.
- TST11 = Test that MPX Register can be loaded and read back.
- TST12 = Tests that all channels can be loaded into MPX register and read back.
- TST13 = Tests auto-Increment mode of MPX register.
- TST14 = Test to see if conversion can be made in specified time.

### 5.2 Miscellaneous Tests

- A. IOT Scope Loop Test - enables IOT to be repeated for troubleshooting.
- B. External Enable Test - utilizes DK8/E Real Time Clock to start conversion, NOTE: This test can be used only if DK8/E is present in system.

- C. Display Converted Value In AC - used to calibrate converter, (See setup procedure of AD8E).
- D. LAB8-E SYSTEM CHECKS - assures reliability of system as homogeneous unit,

### 5.3 Analog Tests

- A. Successive Reads Tests - checks for noise in A-D buffer logic,
- B. Monotonicity Test - checks that all specified values can be converted,
- C. Resolution Accuracy Test - samples a known voltage 64 times and checks that resolution is within specification,
- D. Multiplexer Noise Test - checks for noise in MPX, ENABLE, and STATUS REGISTER,

## 6. ERROR REPORTS

### 6.1 Logic Errors

Message will be typed out once per error on teletype stating test number and nature of failure,

### 6.2 Other Errors

Message will be typed out on teletypewriter stating nature of failure,

## 7. LISTING

/MAINDEC-00-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

/AD8EA,AM8EA,AM8EB

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/DATE REVISED: 6 MAR 72

/REVISED BY: P. T. COYNE

## /IOT DEFINITIONS

|      |       |       |       |                                                          |
|------|-------|-------|-------|----------------------------------------------------------|
| 4520 | ADCL= | JMS I | XADCL | /CLEAR ALL                                               |
| 4521 | ADLM= | JMS I | XADLM | /LOAD MPX REG FROM AC8-11 CLA                            |
| 4522 | ADST= | JMS I | XADST | /CLEAR FLAGS, START CONVERSION                           |
| 4523 | ADRB= | JMS I | XADRB | /CLEAR DONE, READ A-D BUFFER INTO AC                     |
| 4524 | ADSK= | JMS I | XADSK | /SKIP ON A-D DONE, DO NOT CLEAR FLAG                     |
| 4525 | ADSE= | JMS I | XADSE | /SKIP ON TMG ERROR, DO NOT CLEAR FLAG                    |
| 4526 | ADLE= | JMS I | XADLE | /LOAD ENAB REG FROM AC 2-5, CLA                          |
| 4527 | ADRS= | JMS I | XADRS | /READ STATUS, ENAB, MPX REG INTO AC                      |
| 4530 | CLOE= | JMS I | XCLOE | /AC TO CLOCK ENABLE                                      |
| 4531 | CLSK= | JMS I | XCLSK | /SKIP ON CLOCK OVERFLOW                                  |
| 4532 | CLZE= | JMS I | XCLZE | /ONES IN AC CLEAR CLOCK ENABLE REGISTER                  |
| 4533 | CLSA= | JMS I | XCLSA | /CLOCK STATUS TO AC, AC ONES CLEAR CLOCK STATUS REGISTER |
| 4534 | CLED= | JMS I | XCLED | /CLOCK ENABLE TO AC                                      |
| 4535 | CLAB= | JMS I | XCLAB | /AC ONES TO CLOCK BUFFER                                 |
| 4536 | DISD= | JMS I | XDISD | /SKIP ON DISPLAY DONE                                    |
| 4537 | DILX= | JMS I | XDILX | /LOAD X                                                  |
| 4540 | DILY= | JMS I | XDILY | /LOAD Y                                                  |
| 4541 | DIXY= | JMS I | XDIXY | /INTENSIFY                                               |
| 4542 | DILE= | JMS I | XDILE | /LOAD DISPLAY ENABLE FROM AC                             |
| 6007 | CAF=  | 6007  |       |                                                          |
| 7002 | BSW=  | 7002  |       |                                                          |

## /MPX, ENABLE, STATUS REGISTER

/ 0 AD DONE  
 / 1 TIMING ERROR  
 / 2 ENABLE INTERRUPT ON AD DONE  
 / 3 ENABLE INTERRUPT ON TIMING ERROR  
 / 4 ENABLE EXTERNAL AD START  
 / 5 AUTO-INCREMENT MODE  
 / 6,7 NOT USED  
 / 8-11 MPX CHANNEL 0-17 OCTAL

## /STARTING ADDRESS

## TEST

|      |                                      |
|------|--------------------------------------|
| /    |                                      |
| /200 | NORMAL START FOR CONTROL LOGIC TESTS |
| /201 | IOT SCOPE LOOP                       |
| /202 | DISPLAY CONVERTED VALUE IN AC        |
| /203 | EXTERNAL ENABLE TEST                 |
| /204 | MONOTONICITY TEST                    |
| /205 | ACCURACY TEST                        |
| /206 | SUCCESSIVE READS TEST                |
| /207 | MULTIPLEXER NOISE TEST               |
| /210 | SYSTEM CHECK FOR LAB8-E              |

```

0000 0000 *0
0000 0000 0
0001 5402 JMP I ,+1
0002 0000 0
0003 5404 JMP I ,+1
0004 0000 0
0005 7402 HLT

```

```

0017 0017 *17
0017 0145 MSGPNT, ERMMSG

```

```

0020 0020 *20
0020 4000 SW0, 4000 /SWITCH REG 0 INHIBIT TYPEOUT
0021 2000 SW1, 2000 / 1 HALT ON ERROR
0022 1000 SW2, 1000 / 2 SCOPE LOOP OVERRIDE
0023 0400 SW3, 0400 / 3 CALIBRATION TEST HALT
0024 0200 SW4, 0200 / 4 EXTERNAL ENABLE
0025 0100 SW5, 0100 / 5 SELECT TEST
0026 0000 TEMP0, 0 /STORAGE BUFFER 0
0027 0000 TEMP1, 0 /STORAGE BUFFER A
0030 0000 TEMPA, 0 / B
0031 0000 TEMPB, 0 / C
0032 0000 TEMPC, 0 / D
0033 0000 TEMPD, 0 /
0033 0000 CNTR1, 0 /MONOTONICITY COUNTER
0034 0000 TALLY, 0
0035 1226 ERR, ERTYP /ERROR REPORT ROUTINE
0036 1000 XCONVT, CONVT /DISPLAY CONVERTED VALUE
0037 1400 XINSTR, INSTR /IOT SCOPE LOOP
0040 2000 XMONOT, MONOT /MONOTONICITY TEST
0041 0207 K207, 207 /BELL CODE
0042 0212 K212, 212 /LINE FEED
0043 0215 K215, 215 /CARRIAGE RETURN
0044 6500 K6500, 6500
0045 7777 M1, 7777
0046 7776 M2, 7776
0047 7774 M4, 7774
0050 1000 K1000, 1000
0051 0077 K77, 0077
0052 1200 XMOVE, MOVE
0053 1024 EXTBL, EXTL
0054 4377 XSTOR, STORAG=1
0055 2400 XCOMPR, COMPAR
0056 2200 XRESOL, RESOL
0057 2051 XNOISE, NOISE
0060 2103 XGLIT, GLITCH
0061 2600 XSYST, SYST
0062 7777 ERSWIT, 7777
0063 0000 CHAN, 0
0064 1600 TAL, XTAL
0065 1647 SELECT, XSELEC
0066 1552 SETUP, XSETUP
0077 0000 *77
0077 0000 CHNL, 0

```

```

0100 0001 1
01 0002 2

```

|      |      |    |
|------|------|----|
| 0102 | 0003 | 3  |
| 0103 | 0004 | 4  |
| 0104 | 0005 | 5  |
| 0105 | 0006 | 6  |
| 0106 | 0007 | 7  |
| 0107 | 0010 | 10 |
| 0110 | 0011 | 11 |
| 0111 | 0012 | 12 |
| 0112 | 0013 | 13 |
| 0113 | 0014 | 14 |
| 0114 | 0015 | 15 |
| 0115 | 0016 | 16 |
| 0116 | 0017 | 17 |
| 0117 | 0000 | 0  |

0120 \*120

/IOT LINKS

|      |      |        |        |
|------|------|--------|--------|
| 0120 | 1410 | XADCL, | XXADCL |
| 0121 | 1414 | XADLM, | XXADLM |
| 0122 | 1420 | XADST, | XXADST |
| 0123 | 1424 | XADRB, | XXADRB |
| 0124 | 1430 | XADSK, | XXADSK |
| 0125 | 1436 | XADSE, | XXADSE |
| 0126 | 1444 | XADLE, | XXADLE |
| 0127 | 1450 | XADRS, | XXADRS |
| 0130 | 1454 | XCLOE, | XXCLOE |
| 0131 | 1460 | XCLSK, | XXCLSK |
| 0132 | 1466 | XCLZE, | XXCLZE |
| 0133 | 1472 | XCLSA, | XXCLSA |
| 0134 | 1476 | XCLED, | XXCLED |
| 0135 | 1502 | XCLAB, | XXCLAB |
| 0136 | 1506 | XDISD, | XXDISD |
| 0137 | 1514 | XDILX, | XXDILX |
| 0140 | 1520 | XDILY, | XXDILY |
| 0141 | 1524 | XDIXY, | XXDIXY |
| 0142 | 1530 | XDILE, | XXDILE |

0145 \*145

/ERROR MESSAGE LINKS

|      |      |        |        |
|------|------|--------|--------|
| 0145 | 3200 | ERMSG, | EMSG0  |
| 0146 | 3244 |        | EMSG1  |
| 0147 | 3302 |        | EMSG2  |
| 0150 | 3344 |        | EMSG3  |
| 0151 | 3372 |        | EMSG4  |
| 0152 | 3422 |        | EMSG5  |
| 0153 | 3452 |        | EMSG6  |
| 0154 | 3503 |        | EMSG7  |
| 0155 | 3540 |        | EMSG10 |

|      |      |        |
|------|------|--------|
| 0156 | 3601 | EMSG11 |
| 0157 | 3637 | EMSG12 |
| 0160 | 3677 | EMSG13 |
| 0161 | 3744 | EMSG14 |

```

0200      0200      *200
0200      5211      JMP      START      /NORMAL START
0201      5437      JMP I    XINSTR      /IOT SCOPE LOOP OPTION
0202      5436      JMP I    XCONVT      /DISPLAY CONVERTED VALUE OPTION
0203      5453      JMP I    EXTBL      /EXTERNAL ENABLE TEST
0204      5440      JMP I    XMONOT      /MONOTONICITY TEST
0205      5456      JMP I    XRESOL      /ACCURACY TEST
0206      5457      JMP I    XNOISE      /SUCCESSIVE READS TEST
0207      5460      JMP I    XGLIT      /MPX NOISE TEST
0210      5461      JMP I    XSYST      /LABB=E SYSTEM CHECK
0211      7402      START,  HLT
0212      7604      LAS
0213      0025      AND      SW5      /SELECT SPECIFIC TEST?
0214      7440      SZA      /SKIP IF NO
0215      4465      JMS I    SELECT      /YES

/HOUSEKEEPING
0216      7300      INITL,  CLA CLL
0217      4777      JMS      MESSAGE
0220      4161      XLABEL
0221      1376      TAD      (144
0222      3017      DCA      MSGPNT      /INITIALIZE ERROR POINTER
0223      4466      JMS I    SETUP
0224      6007      TST0,  CAF
0225      4524      ADSK
0226      5231      JMP      ,+3      /CHECK FOR DONE FLAG - SHOULD BE CLEARED BY INIT
0227      4435      JMS I    ERR      /DONE FLAG NOT CLEARED
0230      0224      TST0
0231      4525      ADSE      /CHECK FOR TIMING ERROR FLAG - SHOULD BE CLEARED BY INIT
0232      5237      JMP      ,+3
0233      4435      JMS I    ERR      /TIMING ERROR FLAG NOT CLEARED
0234      0224      TST0
0235      5240      JMP      TST1-1
0236      5224      JMP      TST0
0237      4464      JMS I    TAL

/CHECK TO SET DONE FLAG AND CLEAR DONE FLAG
0240      4466      JMS I    SETUP
0241      7200      TST1,  CLA
0242      4522      ADST      /CONVERT, RESULTS NOT NEEDED
0243      1177      TAD      [-100
0244      3026      DCA      TEMP0
0245      2026      ISZ      TEMP0
0246      5245      JMP      ,+1
0247      4524      ADSK
0250      7410      SKP
0251      5235      JMP      ,+4
0252      4435      JMS I    ERR      /FLAG NOT SET
0253      0241      TST1

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0254      5265      JMP      TST2-1
0255      4520      ADCL
0256      4524      ADSK
0257      5264      JMP      ,+5      /CLEAR FLAG
0258      60      4435      JMS I    ERR      /CHECK FOR FLAG
0261      0241      TST1      /FLAG CLEARED

```

|      |      |       |        |  |
|------|------|-------|--------|--|
| 2    | 5265 | JMP   | TST2-1 |  |
| 0263 | 5241 | JMP   | TST1   |  |
| 0264 | 4464 | JMS I | TAL    |  |

/CHECK TO SET TIMING ERROR FLAG AND CLEAR TIMING ERROR FLAG

|      |      |       |        |                                         |
|------|------|-------|--------|-----------------------------------------|
| 0265 | 4466 | JMS I | SETUP  |                                         |
| 0266 | 7200 | TST2, | CLA    |                                         |
| 0267 | 4522 |       | ADST   | /TWO A=D STARTS TO PRODUCE TIMING ERROR |
| 0270 | 4522 |       | ADST   |                                         |
| 0271 | 4525 |       | ADSE   | /CHECK FOR TIMING ERROR FLAG            |
| 0272 | 7410 |       | SKP    |                                         |
| 0273 | 5276 | JMP   | ,+3    |                                         |
| 0274 | 4435 | JMS I | ERR    | /FLAG NOT SET                           |
| 0275 | 0266 | TST2  |        |                                         |
| 0276 | 4520 | ADCL  |        | /CLEAR FLAG                             |
| 0277 | 4525 | ADSE  |        | /CHECK FLAG                             |
| 0300 | 5305 | JMP   | ,+5    |                                         |
| 0301 | 4435 | JMS I | ERR    | /FLAG NOT CLEARED                       |
| 0302 | 0266 | TST2  |        |                                         |
| 0303 | 5306 | JMP   | TST3-1 |                                         |
| 0304 | 5266 | JMP   | TST2   |                                         |
| 0305 | 4464 | JMS I | TAL    |                                         |

/TEST FOR UNEXPECTED INTERRUPT REQUEST

|      |      |        |        |                                |
|------|------|--------|--------|--------------------------------|
| 0306 | 4466 | JMS I  | SETUP  |                                |
| 0307 | 7200 | TST3,  | CLA    |                                |
| 0310 | 1176 |        | TAD    | [TST3S                         |
| 0311 | 3004 |        | DCA    | 4                              |
| 0312 | 1317 |        | TAD    | ,+5                            |
| 0313 | 3001 |        | DCA    | 1                              |
| 0314 | 6001 |        | ION    | /TURN INT ON                   |
| 0315 | 7000 |        | NOP    |                                |
| 0316 | 5322 | JMP    | ,+4    |                                |
| 0317 | 4435 | JMS I  | ERR    | /UNEXPECTED INTERRUPT OCCURRED |
| 0320 | 0307 | TST3   |        |                                |
| 0321 | 5326 | JMP    | TST4-1 |                                |
| 0322 | 6002 | TST3S, | IOF    | /TURN INT OFF                  |
| 0323 | 7410 |        | SKP    |                                |
| 0324 | 5307 | JMP    | TST3   |                                |
| 0325 | 4464 | JMS I  | TAL    |                                |

/TEST THAT ADRB JAM TRANSFERS TO AC

|      |      |       |         |                  |
|------|------|-------|---------|------------------|
| 0326 | 4466 | JMS I | SETUP   |                  |
| 0327 | 7240 | TST4, | CLA CMA | /AC=7777         |
| 0330 | 4523 |       | ADRB    | /SHOULD CLEAR AC |
| 0331 | 3027 |       | DCA     | TEMPA            |
| 0332 | 7040 |       | CMA     |                  |
| 0333 | 4523 |       | ADRB    | /READ WITH AC=0  |

|      |      |       |        |           |
|------|------|-------|--------|-----------|
| 0334 | 7041 | CIA   |        |           |
| 0335 | 1027 | TAD   | TEMPA  |           |
| 0336 | 7440 | SZA   |        | /EQUAL?   |
| 0337 | 7410 | SKP   |        |           |
| 0340 | 5345 | JMP   | ,+5    |           |
| 0341 | 4435 | JMS I | ERR    | /NO-ERROR |
| 0342 | 0327 | TST4  |        |           |
| 0343 | 5346 | JMP   | TST5-1 |           |

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0344 5327 JMP TST4
0345 4464 JMS I TAL

```

/TEST THAT ADRS JAM TRANSFERS TO AC

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0346 4466 JMS I SETUP
0347 4520 TST5, ADCL
0350 4521 ADLM
0351 4522 ADST
0352 4524 ADSK
0353 5352 JMP ,+1
0354 7340 CLA CMA CLL /AC=7777
0355 4527 ADRS
0356 3027 DCA TEMPA /SAVE AC, SHOULD BE 4000
0357 1027 TAD TEMPA
0360 7004 RAL
0361 7440 SZA /DID ADRS CLEAR AC?
0362 7410 SKP
0363 5370 JMP ,+3
0364 4435 JMS I ERR /NO
0365 0347 TST5
0366 5775 JMP TST6=1
0367 5347 JMP TST5
0370 4464 JMS I TAL
0371 5775 JMP TST6=1

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0375 0400
0376 0144
0377 1274
0400

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PAGE

/CHECKS THAT ENABLE REGISTER CAN BE LOADED AND READ BACK

```

0400 4466 JMS I SETUP
0401 7300 TST6, CLA CLL
0402 1175 TAD [17 /GET BITS AND
0403 7002 BSW /PLACE IN AC 2-5
0404 4526 ADLE /LOAD
0405 7440 SZA
0406 7410 SKP
0407 5212 JMP ,+3
0410 4435 JMS I ERR /AC NOT CLEARED BY ADLE
0411 0401 TST6
0412 7040 CMA
0413 4527 ADRS /READ BACK
0414 7002 BSW
0415 1174 TAD [7761 /CHECK FOR ONLY AC 2-5 SET

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/MAINDEC-08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-5

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0416 7440 SZA
0417 7410 SKP
0420 5225 JMP ,+3
0421 4435 JMS I ERR /WRONG BITS
0422 0401 TST6
0423 5226 JMP TST7=1
0424 5201 JMP TST6
0425 4464 JMS I TAL /DONE?

```

/GENERATE INTERRUPT WITH A-D DONE FLAG

|      |      |       |         |                        |
|------|------|-------|---------|------------------------|
| 0426 | 4466 | JMS I | SETUP   |                        |
| 0427 | 7200 | TST7, | CLA     |                        |
| 0430 | 4522 | ADST  |         | /CONVERT               |
| 0431 | 4524 | ADSK  |         | /DONE?                 |
| 0432 | 5231 | JMP   | ,=1     | /WAIT                  |
| 0433 | 1173 | TAD   | [DON1   |                        |
| 0434 | 3002 | DCA   | 2       | /RETURN POINTER        |
| 0435 | 1050 | TAD   | K1000   |                        |
| 0436 | 4526 | ADLE  |         | /LOAD INTERRUPT ENABLE |
| 0437 | 6001 | ION   |         |                        |
| 0440 | 7000 | NOP   |         |                        |
| 0441 | 6002 | IOF   |         |                        |
| 0442 | 4435 | JMS I | ERR     | /DID NOT INTERRUPT     |
| 0443 | 0427 | TST7  |         |                        |
| 0444 | 5231 | JMP   | TST10=1 |                        |
| 0445 | 4520 | DON1, | ADCL    | /CLEAR WORLD           |
| 0446 | 7410 | SKP   |         |                        |
| 0447 | 5227 | JMP   | TST7    |                        |
| 0450 | 4464 | JMS I | TAL     |                        |

/GENERATE INTERRUPT WITH TIMING ERROR FLAG

|      |      |         |         |                        |
|------|------|---------|---------|------------------------|
| 0451 | 4466 | JMS I   | SETUP   |                        |
| 0452 | 7200 | TST10,  | CLA     |                        |
| 0453 | 1172 | TAD     | [TMG1   |                        |
| 0454 | 3002 | DCA     | 2       |                        |
| 0455 | 4522 | ADST    |         | /CAUSE ERROR HERE      |
| 0456 | 4522 | ADST    |         |                        |
| 0457 | 4524 | ADSK    |         | /DONE?                 |
| 0460 | 5237 | JMP     | ,=1     |                        |
| 0461 | 4525 | ADSE    |         | /TIMING ERROR?         |
| 0462 | 5261 | JMP     | ,=1     |                        |
| 0463 | 7300 | CLA CLL |         |                        |
| 0464 | 1050 | TAD     | K1000   |                        |
| 0465 | 7010 | RAR     |         |                        |
| 0466 | 4526 | ADLE    |         | /LOAD INTERRUPT ENABLE |
| 0467 | 6001 | ION     |         | /INT ON                |
| 0470 | 7000 | NOP     |         |                        |
| 0471 | 6002 | IOF     |         | /INT OFF               |
| 0472 | 4435 | JMS I   | ERR     | /DID NOT INTERRUPT     |
| 0473 | 0452 | TST10   |         |                        |
| 0474 | 5301 | JMP     | TST11=1 |                        |
| 0475 | 4520 | TMG1,   | ADCL    | /CLEAR WORLD           |
| 0476 | 7410 | SKP     |         |                        |
| 0477 | 5252 | JMP     | TST10   |                        |
| 0500 | 4464 | JMS I   | TAL     |                        |

/LOAD AND READ MPX REG

|      |      |        |         |                             |
|------|------|--------|---------|-----------------------------|
| 0501 | 4466 | JMS I  | SETUP   |                             |
| 0502 | 7240 | TST11, | CLA CMA |                             |
| 0503 | 4521 | ADLM   |         |                             |
| 0504 | 7450 | SNA    |         | /CHECK IF AC CLEARED        |
| 0505 | 5311 | JMP    | ,+4     |                             |
| 0506 | 4435 | JMS I  | ERR     | /AC WAS NOT CLEARED BY ADLM |
| 0507 | 0502 | TST11  |         |                             |
| 0510 | 7200 | CLA    |         |                             |

|      |      |       |         |                       |
|------|------|-------|---------|-----------------------|
| 0511 | 4521 | ADLM  |         | /LOAD MPX REG WITH 00 |
| 0512 | 4527 | ADRS  |         | /READ MPX REG         |
| 0513 | 0175 | AND   | [17     | /MASK FOR MPX REG     |
| 0514 | 7440 | SEA   |         |                       |
| 0515 | 7410 | SKP   |         |                       |
| 0516 | 5321 | JMP   | ,+3     |                       |
| 0517 | 4435 | JMS I | ERR     | /MPX REG NOT 0        |
| 0520 | 0502 | TST11 |         |                       |
| 0521 | 7040 | CMA   |         |                       |
| 0522 | 0175 | AND   | [17     |                       |
| 0523 | 4521 | ADLM  |         | /MPX REG SET TO 17    |
| 0524 | 4527 | ADRS  |         | /READ MPX REG         |
| 0525 | 0175 | AND   | [17     |                       |
| 0526 | 1171 | TAD   | [7760   | /MASK                 |
| 0527 | 7040 | CMA   |         |                       |
| 0530 | 7440 | SEA   |         |                       |
| 0531 | 7410 | SKP   |         |                       |
| 0532 | 5337 | JMP   | ,+3     |                       |
| 0533 | 4435 | JMS I | ERR     | /MPX REG NOT 17       |
| 0534 | 0502 | TST11 |         |                       |
| 0535 | 5777 | JMP   | TST12=1 |                       |
| 0536 | 5302 | JMP   | TST11   |                       |
| 0537 | 4464 | JMS I | TAL     |                       |
| 0540 | 5777 | JMP   | TST12=1 |                       |

0577 0600  
0600 PAGE

/LOAD MPX REG WITH EACH CHANNEL

|      |      |        |         |                       |
|------|------|--------|---------|-----------------------|
| 0600 | 4466 | JMS I  | SETUP   |                       |
| 0601 | 7300 | TST12, | CLA CLL |                       |
| 0602 | 3026 | DCA    | TEMP0   |                       |
| 0603 | 1026 | TAD    | TEMP0   | /CHANNEL INTO AC      |
| 0604 | 7040 | CMA    |         |                       |
| 0605 | 3027 | DCA    | TEMPA   | /COMPLEMENTED CHANNEL |
| 0606 | 1026 | TAD    | TEMP0   |                       |
| 0607 | 4521 | ADLM   |         | /LOAD IT              |
| 0610 | 4527 | ADRS   |         | /READ MPX REG         |
| 0611 | 0175 | AND    | [17     | /MASK 8-11            |
| 0612 | 3030 | DCA    | TEMPB   | /STORE IT             |
| 0613 | 1027 | TAD    | TEMPA   | /CHECK IT             |
| 0614 | 1030 | TAD    | TEMPB   |                       |
| 0615 | 7001 | IAC    |         |                       |

/MAINDEC=08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-7

|      |      |         |         |                          |
|------|------|---------|---------|--------------------------|
| 0616 | 7440 | SEA     |         |                          |
| 0617 | 7410 | SKP     |         |                          |
| 0620 | 5224 | JMP     | ,+4     |                          |
| 0621 | 4435 | JMS I   | ERR     | /WRONG CHANNEL           |
| 0622 | 0601 | TST12   |         |                          |
| 0623 | 5236 | JMP     | TST13=1 |                          |
| 0624 | 1171 | TAD     | [7760   |                          |
| 0625 | 1026 | TAD     | TEMP0   |                          |
| 0626 | 7001 | IAC     |         |                          |
| 0627 | 7440 | SEA     |         | /DONE WITH ALL CHANNELS? |
| 0630 | 7410 | SKP     |         | /NO                      |
| 0631 | 5235 | JMP     | ,+4     | /YES                     |
| 0632 | 7300 | CLA CLL |         |                          |
| 0633 | 2026 | ISE     | TEMP0   | /SET N CHANNEL           |

```

034 5203      JMP      TST12+2
0635 4464      JMS I    TAL              /DONE WITH TEST?

/AUTO-INCREMENT MODE TEST
0636 4466      JMS I    SETUP
0637 7300      TST13,   CLA CLL
0640 1170      TAD      [76
0641 3010      DCA      10
0642 4520      ADCL
0643 7200      AUTO1,   CLA
0644 1410      TAD I    10              /CHANNEL N
0645 3027      DCA      TEMPB
0646 1027      TAD      TEMPB
0647 7040      CMA
0650 3030      DCA      TEMPB
0651 1025      TAD      SW5              /AUTO-INCREMENT BIT
0652 4526      ADLE              /LOAD ENABLE REG
0653 1027      TAD      TEMPB              /CHANNEL N
0654 4521      ADLM              /LOAD MPX REG
0655 4522      ADST              /START CONVERSION
0656 4524      ADSK              /WAIT FOR
0657 5256      JMP      ,=1              /DONE FLAG
0660 4527      ADRS              /READ STATUS
0661 0175      AND      [17              /MASK OUT ALL BUT MPX REG
0662 3031      DCA      TEMPB
0663 1027      TAD      TEMPB              /CHECK IF CHANNEL I7 INCREMENTED TO 0
0664 1174      TAD      [7761
0665 7640      SZA CLA              /IF CHANNEL I7 SKIP
0666 5272      JMP      ,+4
0667 1410      TAD I    10
0670 1031      TAD      TEMPB
0671 5274      JMP      AUTO2              /AC SHOULD = 0
0672 1031      TAD      TEMPB
0673 1030      TAD      TEMPB              /CHECK FOR CHANNEL N+1
0674 7440      AUTO2,   SZA
0675 7410      SKP
0676 5302      JMP      ,+4
0677 4435      JMS I    ERR              /WRONG CHANNEL
0700 0637      TST13
0701 5310      JMP      TST14-1

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/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10 V141

21-MAR-72

13125

PAGE 2-8

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0702 1031      TAD      TEMPB
0703 7440      SZA              /LAST CHANNEL?
0704 5243      JMP      AUTO1          /NO
0705 7410      SKP
0706 5237      JMP      TST13
0707 4464      JMS I    TAL

```

/ROUTINE TO CHECK THAT CONVERSION CAN BE MADE IN 20 MICROSECS

```

0710 4466      JMS I    SETUP
0711 7300      TST14,   CLA CLL
0712 1377      TAD      (-6
0713 3026      DCA      TEMPB
0714 4520      ADCL
0715 4522      ADST
0716 2026      ISZ      TEMPB

```

|      |      |            |         |                                      |
|------|------|------------|---------|--------------------------------------|
| 0717 | 5316 | JMP        | ,-1     |                                      |
| 0720 | 4524 | ADSK       |         |                                      |
| 0721 | 7410 | SKP        |         |                                      |
| 0722 | 5327 | JMP        | ,+5     |                                      |
| 0723 | 4435 | JMS I      | ERR     | /TIME OUT ERROR                      |
| 0724 | 0711 | TST14      |         |                                      |
| 0725 | 5330 | JMP        | FINIS   |                                      |
| 0726 | 5311 | JMP        | TST14   |                                      |
| 0727 | 4464 | JMS I      | TAL     |                                      |
| 0730 | 7604 | FINIS, LAS |         |                                      |
| 0731 | 0020 | AND        | SW0     | /SWITCH SET TO DELETE                |
| 0732 | 7640 | SZA CLA    |         | /TYPEOUT OF END LOGIC TEST           |
| 0733 | 5337 | JMP        | ,+4     |                                      |
| 0734 | 4776 | JMS        | MESSAGE |                                      |
| 0735 | 4146 | XEND       |         |                                      |
| 0736 | 7200 | CLA        |         |                                      |
| 0737 | 1041 | TAD        | K207    |                                      |
| 0740 | 4775 | JMS        | PRLP    |                                      |
| 0741 | 5774 | JMP        | TST0-3  | /RETURN TO BEGINNING OF LOGIC TESTS. |

0774 0221  
0775 1534  
0776 1274  
0777 7772  
1000

PAGE

/ROUTINE TO DISPLAY CONVERTED VALUE IN AC.

|      |      |             |       |                   |
|------|------|-------------|-------|-------------------|
| 1000 | 4520 | CONVT, ADCL |       | /CLEAR WORLD      |
| 1001 | 3026 | DCA         | TEMP0 |                   |
| 1002 | 7604 | LAS         |       | /LOAD CHANNEL     |
| 1003 | 4521 | ADLM        |       | /LOAD MPX REG     |
| 1004 | 4522 | ADST        |       | /CONVERT          |
| 1005 | 4524 | ADSK        |       | /DONE?            |
| 1006 | 5205 | JMP         | ,-1   | /WAIT             |
| 1007 | 4523 | ADRB        |       | /READ A-D BUFFER  |
| 1010 | 2026 | ISZ         | TEMP0 | /STALL TO DISPLAY |
| 1011 | 5210 | JMP         | ,-1   | /CONVERTED VALUE  |
| 1012 | 2026 | ISZ         | TEMP0 | /IN AC FOR        |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-9

|      |      |         |       |                                       |
|------|------|---------|-------|---------------------------------------|
| 1013 | 5212 | JMP     | ,-1   | /33 MILLISECONDS                      |
| 1014 | 3031 | DCA     | TEMPC |                                       |
| 1015 | 7604 | LAS     |       | /CHECK IF HALT DESIRED                |
| 1016 | 0023 | AND     | SW3   |                                       |
| 1017 | 7650 | SNA CLA |       |                                       |
| 1020 | 5223 | JMP     | ,+3   |                                       |
| 1021 | 1031 | TAD     | TEMPC |                                       |
| 1022 | 7402 | HLT     |       | /PRESS CONTINUE IF NOT DONE ADJUSTING |
| 1023 | 5200 | JMP     | CONVT | /LOOP                                 |

/ROUTINE TO CHECK FOR EXTERNAL ENABLE FROM REAL TIME CLOCK

|      |      |             |       |                            |
|------|------|-------------|-------|----------------------------|
| 1024 | 4466 | EXTL, JMS I | SETUP |                            |
| 1025 | 4520 | ADCL        |       | /CLEAR ALL                 |
| 1026 | 7604 | LAS         |       |                            |
| 1027 | 0024 | AND         | SW4   | /CHECK EXT'L ENABLE SWITCH |

|      |      |         |       |                                |
|------|------|---------|-------|--------------------------------|
| 1030 | 7450 | SNA     |       |                                |
| 1031 | 7402 | HLT     |       | /SWITCH NOT SET, STOP!         |
| 1032 | 7604 | LAS     |       |                                |
| 1033 | 0024 | AND     | SW4   |                                |
| 1034 | 4526 | ADLE    |       | /LOAD EXTERNAL ENABLE INTO ADC |
| 1035 | 7604 | LAS     |       |                                |
| 1036 | 0175 | AND     | [17   |                                |
| 1037 | 4521 | ADLM    |       | /LOAD CHANNEL FROM SW8=11      |
| 1040 | 1377 | TAD     | (4340 | /LOAD CLOCK ENABLE REG         |
| 1041 | 4530 | CLOE    |       | /TRIGGER FROM RTC              |
| 1042 | 7040 | CMA     |       |                                |
| 1043 | 4532 | CLZE    |       |                                |
| 1044 | 4531 | CLSK    |       | /OCCURS ON OVERFLOW            |
| 1045 | 5244 | JMP     | ,=1   |                                |
| 1046 | 4533 | CLSA    |       | /STOP CLOCK                    |
| 1047 | 7240 | CLA CMA |       |                                |
| 1050 | 4532 | CLZE    |       |                                |
| 1051 | 7200 | CLA     |       |                                |
| 1052 | 2026 | ISZ     | TEMP0 | /TIME OUT                      |
| 1053 | 5232 | JMP     | ,=1   |                                |
| 1054 | 4524 | ADSK    |       |                                |
| 1055 | 4776 | JMS     | ERPT3 | /CONVERSION NOT MADE           |
| 1056 | 4523 | ADRB    |       |                                |
| 1057 | 3027 | DCA     | TEMPA | /STORE CONVERSION              |
| 1060 | 7604 | LAS     |       |                                |
| 1061 | 0022 | AND     | SW2   | /LOOP?                         |
| 1062 | 7630 | SNA CLA |       |                                |
| 1063 | 5266 | JMP     | EXTTE | /YES                           |
| 1064 | 1027 | TAD     | TEMPA | /HALT WITH CONVERTED           |
| 1065 | 7402 | HLT     |       | /VALUE IN AC                   |
| 1066 | 4466 | JMS I   | SETUP |                                |
| 1067 | 4520 | ADCL    |       |                                |
| 1070 | 7604 | LAS     |       |                                |
| 1071 | 0024 | AND     | SW4   |                                |
| 1072 | 4526 | ADLE    |       |                                |
| 1073 | 7240 | CLA CMA |       |                                |
| 1074 | 4535 | CLAB    |       | /CLOCK BUFFER = 7777           |

/MAINDEC=08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-10

|      |      |         |       |                                     |
|------|------|---------|-------|-------------------------------------|
| 1075 | 7200 | CLA     |       |                                     |
| 1076 | 1375 | TAD     | (1640 | /TO GIVE TIMING ERROR ON NEXT CLOCK |
| 1077 | 4530 | CLOE    |       |                                     |
| 1100 | 7000 | NOP     |       |                                     |
| 1101 | 4525 | ADSE    |       | /TIMING ERROR SEEN HERE             |
| 1102 | 4776 | JMS     | ERPT3 | /DID NOT RAISE FLAG                 |
| 1103 | 7240 | CLA CMA |       |                                     |
| 1104 | 4532 | CLZE    |       | /CLEAR CLOCK ENABLE REG             |
| 1105 | 7200 | CLA     |       |                                     |
| 1106 | 4520 | ADCL    |       |                                     |
| 1107 | 1024 | TAD     | SW4   |                                     |
| 1110 | 4526 | ADLE    |       | /LOAD EXT/L ENABLE INTO ADC         |
| 1111 | 7240 | CLA CMA |       |                                     |
| 1112 | 4535 | CLAB    |       | /SET THEN CLEAR                     |
| 1113 | 7200 | CLA     |       | /CLOCK BUFFER TO CHECK              |
| 1114 | 4535 | CLAB    |       | /FOR ERRONEOUS START PULSE          |
| 1115 | 1374 | TAD     | (=6   |                                     |
| 1116 | 3031 | DCA     | TEMPC |                                     |
| 1117 | 2031 | ISZ     | TEMPC |                                     |
| 1120 | 5212 | JMP     |       |                                     |

|      |      |      |       |                |
|------|------|------|-------|----------------|
| 1121 | 4524 | ADSK |       | /IF FLAG FOUND |
| 1122 | 5325 | JMP  | ,+3   | /REPORT        |
| 1123 | 4776 | JMS  | ERPT3 | /ERROR         |
| 1124 | 4520 | ADCL |       |                |
| 1125 | 7200 | CLA  |       |                |
| 1126 | 1041 | TAD  | K207  |                |
| 1127 | 4773 | JMS  | PRLP  |                |
| 1130 | 5232 | JMP  | EXT1  |                |

1173 1534  
1174 7772  
1175 1640  
1176 1732  
1177 4340  
1200

PAGE

/SUBROUTINE TO MOVE VARIABLE LENGTH DATA FIELDS

|      |      |        |         |       |
|------|------|--------|---------|-------|
| 1200 | 0000 | MOVE,  | 0       |       |
| 1201 | 7300 |        | CLA CLL |       |
| 1202 | 1600 |        | TAD I   | MOVE  |
| 1203 | 3223 |        | DCA     | FADDR |
| 1204 | 2200 |        | ISZ     | MOVE  |
| 1205 | 1600 |        | TAD I   | MOVE  |
| 1206 | 3224 |        | DCA     | TADDR |
| 1207 | 2200 |        | ISZ     | MOVE  |
| 1210 | 1600 |        | TAD I   | MOVE  |
| 1211 | 3225 |        | DCA     | MCTR  |
| 1212 | 2200 |        | ISZ     | MOVE  |
| 1213 | 7200 | MOVEA, | CLA     |       |
| 1214 | 1623 |        | TAD I   | FADDR |
| 1215 | 3624 |        | DCA I   | TADDR |
| 1216 | 2223 |        | ISZ     | FADDR |

/GET "FROM ADDR" AND  
/STORE  
/GET "TO ADDR" AND  
/STORE  
/GET "MOVE COUNT" AND  
/STORE  
/SETUP FOR EXIT  
/GET "FROM" WORD  
/STORE AT "TO" LOCATION  
/+1 TO "FROM" ADDR

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10 V141

21-MAR-72

13125 PAGE 2-11

|      |      |        |       |       |                   |
|------|------|--------|-------|-------|-------------------|
| 1217 | 2224 |        | ISZ   | TADDR | /+1 TO "TO" ADDR  |
| 1220 | 2225 |        | ISZ   | MCTR  | /ALL WORDS MOVED? |
| 1221 | 5213 |        | JMP   | MOVEA | /NO, RETURN       |
| 1222 | 5600 |        | JMP I | MOVE  | /YES, EXIT        |
| 1223 | 0000 | FADDR, | 0     |       |                   |
| 1224 | 0000 | TADDR, | 0     |       |                   |
| 1225 | 0000 | MCTR,  | 0     |       |                   |

/ERROR TYPEOUT ROUTINE

|      |      |        |         |        |
|------|------|--------|---------|--------|
| 1226 | 0000 | ERTYP, | 0       |        |
| 1227 | 7200 |        | CLA     |        |
| 1230 | 1346 |        | TAD     | IND    |
| 1231 | 7640 |        | SZA CLA |        |
| 1232 | 5243 |        | JMP     | EOUT+1 |
| 1233 | 7604 |        | LAS     |        |
| 1234 | 0020 |        | AND     | SW0    |
| 1235 | 7710 |        | SPA CLA |        |
| 1236 | 5247 |        | JMP     | EOUT+5 |
| 37   | 1417 |        | TAD I   | MSGPNT |

/TYPE ERROR MESSAGE ONE TIME ONLY  
/SUPPRESS TYPEOUT?  
/YES  
/GET POINTER FOR ERROR MESSAGE

|      |      |       |         |                              |
|------|------|-------|---------|------------------------------|
| 40   | 3242 | DCA   | EOUT    |                              |
| 1241 | 4274 | JMS   | MESSAGE |                              |
| 1242 | 7402 | HLT   |         |                              |
| 1243 | 7200 | CLA   |         |                              |
| 1244 | 1346 | TAD   | IND     |                              |
| 1245 | 7640 | SZA   | CLA     |                              |
| 1246 | 5250 | JMP   | ,+2     |                              |
| 1247 | 2346 | ISZ   | IND     |                              |
| 1250 | 7604 | LAS   |         |                              |
| 1251 | 0021 | AND   | SW1     | /HALT ON ERROR SWITCH ON?    |
| 1252 | 7650 | SNA   | CLA     | /SKIP IF ON                  |
| 1253 | 5257 | JMP   | SCOPE   |                              |
| 1254 | 1226 | TAD   | ERTYP   |                              |
| 1255 | 1045 | TAD   | M1      |                              |
| 1256 | 7402 | HLT   |         | /HALT WITH ERROR P.C. IN AC. |
| 1257 | 7604 | LAS   |         |                              |
| 1260 | 0022 | AND   | SW2     | /OVERRIDE LOOP?              |
| 1261 | 7640 | SZA   | CLA     |                              |
| 1262 | 5272 | JMP   | ,+10    |                              |
| 1263 | 1626 | TAD I | ERTYP   | /NO                          |
| 1264 | 3271 | DCA   | EXIT    |                              |
| 1265 | 1017 | TAD   | MSGPNT  |                              |
| 1266 | 1045 | TAD   | M1      |                              |
| 1267 | 3017 | DCA   | MSGPNT  |                              |
| 1270 | 5671 | JMP I | EXIT    |                              |
| 1271 | 7402 | HLT   |         |                              |
| 1272 | 2226 | ISZ   | ERTYP   | /YES                         |
| 1273 | 5626 | JMP I | ERTYP   |                              |

/MESSAGE ROUTINE FOR LOGIC ERRORS

1274 0000 MESSAGE, 0

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-12

|      |      |           |           |
|------|------|-----------|-----------|
| 1275 | 7240 | CLA       | CMA       |
| 1276 | 1674 | TAD I     | MESSAGE   |
| 1277 | 3010 | DCA       | 10        |
| 1300 | 2274 | ISZ       | MESSAGE   |
| 1301 | 1410 | TAD I     | 10        |
| 1302 | 3313 | DCA       | MSRGHT    |
| 1303 | 1313 | TAD       | MSRGHT    |
| 1304 | 7012 | RTR       |           |
| 1305 | 7012 | RTR       |           |
| 1306 | 7012 | RTR       |           |
| 1307 | 4314 | JMS       | TYPECH    |
| 1310 | 1313 | TAD       | MSRGHT    |
| 1311 | 4314 | JMS       | TYPECH    |
| 1312 | 5301 | JMP       | MESSAGE+5 |
| 1313 | 0000 | MSRGHT, 0 |           |
| 1314 | 0000 | TYPECH, 0 |           |
| 1315 | 0051 | AND       | K77       |
| 1316 | 7450 | SNA       |           |
| 1317 | 5674 | JMP I     | MESSAGE   |
| 1320 | 1377 | TAD       | (-40      |
| 1321 | 7510 | SPA       |           |
| 1322 | 5325 | JMP       | ,+3       |
| 1323 | 1376 | TAD       | (240      |
| 1324 | 5340 | JMP       | MTP       |

|      |      |       |        |
|------|------|-------|--------|
| 1325 | 7001 | IAC   |        |
| 1326 | 7440 | SZA   |        |
| 1327 | 5332 | JMP   | ,+3    |
| 1330 | 1043 | TAD   | K215   |
| 1331 | 5340 | JMP   | MTP    |
| 1332 | 7001 | IAC   |        |
| 1333 | 7440 | SZA   |        |
| 1334 | 5337 | JMP   | ,+3    |
| 1335 | 1042 | TAD   | K212   |
| 1336 | 5340 | JMP   | MTP    |
| 1337 | 1375 | TAD   | (336   |
| 1340 | 6046 | MTP,  | TLS    |
| 1341 | 6041 | TSP   |        |
| 1342 | 5341 | JMP   | ,=1    |
| 1343 | 6042 | TCF   |        |
| 1344 | 7200 | CLA   |        |
| 1345 | 5714 | JMP I | TYPECH |
| 1346 | 0000 | IND,  | 0      |

|      |      |                            |                          |
|------|------|----------------------------|--------------------------|
| 1375 | 0336 |                            |                          |
| 1376 | 0240 |                            |                          |
| 1377 | 7740 |                            |                          |
|      | 1400 | PAGE                       |                          |
|      |      | /SCOPE LOOP FOR IOTS 65XX, |                          |
| 1400 | 7000 | INSTR,                     | NOP                      |
| 1401 | 7604 | LAS                        | /SELECT IOT FROM SR 6-11 |
| 1402 | 0051 | AND                        | K77 /MASK OUT AC 0-5     |
| 1403 | 1044 | TAD                        | K6500 /CREATE IOT        |
| 1404 | 3205 | DCA                        | ,+1                      |
| 1405 | 7402 | HLT                        | /LOCATION OF IOT         |

/MAINDEC=00-BHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC      PAL10    V141    21-MAR-72    13125    PAGE 2-13

|      |      |     |                |
|------|------|-----|----------------|
| 1406 | 7000 | NOP | /POSSIBLE SKIP |
| 1407 | 5201 | JMP | INSTR+1 /LOOP  |

/IOT SUBROUTINES

|      |      |           |                   |
|------|------|-----------|-------------------|
| 1410 | 0000 | XXADCL, 0 |                   |
| 1411 | 6530 | 6530      | /CLEAR ALL        |
| 1412 | 5610 | JMP I     | XXADCL            |
| 1413 | 7402 | HLT       |                   |
| 1414 | 0000 | XXADLM, 0 |                   |
| 1415 | 6531 | 6531      | /LOAD MPX REG     |
| 1416 | 5614 | JMP I     | XXADLM            |
| 1417 | 7402 | HLT       |                   |
| 1420 | 0000 | XXADST, 0 |                   |
| 1421 | 6532 | 6532      | /START CONVERSION |
| 1422 | 5620 | JMP I     | XXADST            |
| 1423 | 7402 | HLT       |                   |
| 1424 | 0000 | XXADRB, 0 |                   |
| 1425 | 6533 | 6533      | /READ A-D BUFFER  |
| 1426 | 5624 | JMP I     | XXADRB            |
| 1427 | 7402 | HLT       |                   |

|      |      |           |                       |
|------|------|-----------|-----------------------|
| 1430 | 0000 | XXADSK, 0 |                       |
| 1431 | 6534 | 6534      | /SKIP ON A/D DONE     |
| 1432 | 7410 | SKP       |                       |
| 1433 | 2230 | ISZ       | XXADSK                |
| 1434 | 5630 | JMP I     | XXADSK                |
| 1435 | 7402 | HLT       |                       |
|      |      |           |                       |
| 1436 | 0000 | XXADSE, 0 |                       |
| 1437 | 6535 | 6535      | /SKIP ON TIMING ERROR |
| 1440 | 7410 | SKP       |                       |
| 1441 | 2236 | ISZ       | XXADSE                |
| 1442 | 5636 | JMP I     | XXADSE                |
| 1443 | 7402 | HLT       |                       |
|      |      |           |                       |
| 1444 | 0000 | XXADLE, 0 |                       |
| 1445 | 6536 | 6536      | /LOAD ENABLE REGISTER |
| 1446 | 5644 | JMP I     | XXADLE                |
| 1447 | 7402 | HLT       |                       |
|      |      |           |                       |
| 1450 | 0000 | XXADRS, 0 |                       |
| 1451 | 6537 | 6537      | /READ STATUS REGISTER |
| 1452 | 5650 | JMP I     | XXADRS                |
| 1453 | 7402 | HLT       |                       |
|      |      |           |                       |
| 1454 | 0000 | XXCLOE, 0 |                       |
| 1455 | 6132 | 6132      | /LOAD CLOCK ENABLE    |
| 1456 | 5654 | JMP I     | XXCLOE                |
| 1457 | 7402 | HLT       |                       |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-14

|      |      |           |                                                 |
|------|------|-----------|-------------------------------------------------|
| 1460 | 0000 | XXCLSK, 0 |                                                 |
| 1461 | 6131 | 6131      | /SKIP ON CLOCK OVERFLOW                         |
| 1462 | 7410 | SKP       |                                                 |
| 1463 | 2260 | ISZ       | XXCLSK                                          |
| 1464 | 5660 | JMP I     | XXCLSK                                          |
| 1465 | 7402 | HLT       |                                                 |
|      |      |           |                                                 |
| 1466 | 0000 | XXCLZE, 0 |                                                 |
| 1467 | 6130 | 6130      | /ONES IN AC CLEAR CLOCK ENABLE REG              |
| 1470 | 5666 | JMP I     | XXCLZE                                          |
| 1471 | 7402 | HLT       |                                                 |
|      |      |           |                                                 |
| 1472 | 0000 | XXCLSA, 0 |                                                 |
| 1473 | 6135 | 6135      | /CLOCK STATUS TO AC, AC ONES CLR CLK STATUS REG |
| 1474 | 5672 | JMP I     | XXCLSA                                          |
| 1475 | 7402 | HLT       |                                                 |
|      |      |           |                                                 |
| 1476 | 0000 | XXCLED, 0 |                                                 |
| 1477 | 6134 | 6134      | /CLOCK ENABLE TO AC                             |
| 1500 | 5676 | JMP I     | XXCLED                                          |
| 1501 | 7402 | HLT       |                                                 |
|      |      |           |                                                 |
| 1502 | 0000 | XXCLAB, 0 |                                                 |
| 1503 | 6133 | 6133      | /AC ONES TO CLOCK BUFFER                        |
| 1504 | 5702 | JMP I     | XXCLAB                                          |
| 1505 | 7402 | HLT       |                                                 |

|      |      |           |                                |
|------|------|-----------|--------------------------------|
| 1506 | 0000 | XXDISD, 0 |                                |
| 1507 | 6052 | 6052      | /SKIP ON DISPLAY DONE          |
| 1510 | 7410 | SKP       |                                |
| 1511 | 2306 | ISZ       | XXDISD                         |
| 1512 | 5706 | JMP I     | XXDISD                         |
| 1513 | 7402 | HLT       |                                |
|      |      |           |                                |
| 1514 | 0000 | XXDILX, 0 |                                |
| 1515 | 6053 | 6053      | /LOAD X                        |
| 1516 | 5714 | JMP I     | XXDILX                         |
| 1517 | 7402 | HLT       |                                |
|      |      |           |                                |
| 1520 | 0000 | XXDILY, 0 |                                |
| 1521 | 6054 | 6054      | /LOAD Y                        |
| 1522 | 5720 | JMP I     | XXDILY                         |
| 1523 | 7402 | HLT       |                                |
|      |      |           |                                |
| 1524 | 0000 | XXDIXY, 0 |                                |
| 1525 | 6055 | 6055      | /INTENSIFY                     |
| 1526 | 5724 | JMP I     | XXDIXY                         |
| 1527 | 7402 | HLT       |                                |
|      |      |           |                                |
| 1530 | 0000 | XXDILE, 0 |                                |
| 1531 | 6056 | 6056      | /LOAD ENABLE FROM AC, CLEAR AC |
| 1532 | 5730 | JMP I     | XXDILE                         |
| 1533 | 7402 | HLT       |                                |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-15

|                                    |      |         |                            |
|------------------------------------|------|---------|----------------------------|
| /PRINT ROUTINE                     |      |         |                            |
| 1534                               | 0000 | PRLP, 0 |                            |
| 1535                               | 6046 | TLS     | /XMIT CHARACTER            |
| 1536                               | 6041 | TSF     | /WAIT FOR FLAG             |
| 1537                               | 5336 | JMP     | ,=1                        |
| 1540                               | 7200 | CLA     |                            |
| 1541                               | 5734 | JMP I   | PRLP /RETURN               |
|                                    |      |         |                            |
| /CARRIAGE RETURN LINE FEED ROUTINE |      |         |                            |
| 1542                               | 0000 | CRLF, 0 |                            |
| 1543                               | 7240 | CLA CMA |                            |
| 1544                               | 0043 | AND     | K215 /CARRIAGE RETURN CODE |
| 1545                               | 4334 | JMS     | PRLP /PRINT ROUTINE        |
| 1546                               | 7240 | CLA CMA |                            |
| 1547                               | 0042 | AND     | K212 /LINE FEED CODE       |
| 1550                               | 4334 | JMS     | PRLP /PRINT ROUTINE        |
| 1551                               | 5742 | JMP I   | CRLF /RETURN               |

/ROUTINE TO CLEAR WORKING BUFFERS PRIOR TO TEST

|      |      |           |                        |
|------|------|-----------|------------------------|
| 1552 | 0000 | XSETUP, 0 |                        |
| 1553 | 4452 | JMS I     | XMOVE /CLEAR WORK AREA |
| 1554 | 0026 | TEMP0     |                        |
| 1555 | 0027 | TEMPA     |                        |
| 1556 | 7773 | =5        |                        |
| 1557 | 6002 | IOF       |                        |
| 1558 | 6007 | CAF       |                        |

```

      2 0000      ERPT3, 0
1733 7604      LAS
1734 0020      AND      SW0
1735 7710      SPA CLA
1736 5342      JMP      ,+4
1737 4777'     JMS      MESSAGE
1740 4056      MSG22
1741 4776'     JMS      CRLF
1742 7604      LAS
1743 0021      AND      SW1
1744 7650      SNA CLA
1745 5732      JMP      ERPT3
1746 1332      TAD      ERPT3
1747 1045      TAD      M1
1750 7402      HLT

1791 0000      ERPT4, 0
1792 4777'     JMS      MESSAGE
1793 4105      MSG23
1794 4776'     JMS      CRLF

```

/MAINDEC-00-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-18

```

1795 5791      JMP      ERPT4

1756 0000      ERPT5, 0
1757 7604      LAS
1758 0020      AND      SW0
1761 7710      SPA CLA
1762 5366      JMP      ,+4
1763 4777'     JMS      MESSAGE
1764 4122      MSG24
1765 4776'     JMS      CRLF
1766 5756      JMP      ERPT5

```

```

1773 2051
1774 2200
1775 3000
1776 1542
1777 1274
      2000

```

PAGE  
/MONOTONICITY TEST  
MONOT, CLA CLL

```

2000 7300
2001 3027      DCA      TEMP A      /CLEAR N AND
2002 3030      DCA      TEMP B      /N+1 CONVERSION STORAGE
2003 4520      ADCL      /CLEAR CONVERTER
2004 4522      ADST      /START CONVERSION
2005 4524      ADSK      /WAIT FOR DONE
2006 5205      JMP      ,=1
2007 4523      ADRB      /READ A-D BUFFER
2010 3027      DCA      TEMP A      /STORE NTH CONVERSION
2011 7604      LAS      /GET SWITCHES
2012 7040      CMA      /COMPLEMENT FOR DOWN COUNT
2013 3033      DCA      CNTR1
2014 4522      ADST      /DO N+1ST CONVERSION
2015 4524      ADSK

```

CONT,

|      |      |         |       |                                         |
|------|------|---------|-------|-----------------------------------------|
| 2016 | 5215 | JMP     | ,=1   |                                         |
| 2017 | 4523 | ADRB    |       |                                         |
| 2020 | 3030 | DCA     | TEMPB | /SAVE                                   |
| 2021 | 1027 | TAD     | TEMPA | /SUBTRACT                               |
| 2022 | 7041 | CIA     |       |                                         |
| 2023 | 1030 | TAD     | TEMPB |                                         |
| 2024 | 7510 | SPA     |       | />0?                                    |
| 2025 | 7041 | CIA     |       | /NO, TAKE ABSOLUTE VALUE                |
| 2026 | 7450 | SNA     |       | /DIFFERENCE 0?                          |
| 2027 | 5243 | JMP     | OK    | /YES, OK,                               |
| 2030 | 1045 | TAD     | M1    |                                         |
| 2031 | 7650 | SNA CLA |       | /DIFFERENCE = 1?                        |
| 2032 | 5243 | JMP     | OK    | /YES, OK,                               |
| 2033 | 4777 | JMS     | ERPT4 |                                         |
| 2034 | 7200 | CLA     |       |                                         |
| 2035 | 1027 | TAD     | TEMPA | /DIFFERENCE > 1, DISPLAY NYH CONVERSION |
| 2036 | 7402 | HLT     |       |                                         |
| 2037 | 7300 | CLA CLL |       |                                         |

/MAINDEC=08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC PAL10 V141 21-MAR-72 13125 PAGE 2-19

|      |      |         |       |                         |
|------|------|---------|-------|-------------------------|
| 2040 | 1030 | TAD     | TEMPB | /DISPLAY N+1 CONVERSION |
| 2041 | 7402 | HLT     |       |                         |
| 2042 | 5200 | JMP     | MONOT | /RESTART TO RESYNC      |
| 2043 | 2033 | ISZ     | CNTR1 | /STALL                  |
| 2044 | 5243 | JMP     | ,=1   |                         |
| 2045 | 7300 | CLA CLL |       |                         |
| 2046 | 1030 | TAD     | TEMPB | /N+1 CONVERSION BECOMES |
| 2047 | 3027 | DCA     | TEMPA | /N                      |
| 2050 | 5211 | JMP     | CONT  | /GET N+1 CONVERSION     |

/ROUTINE TO TEST FOR EQUALITY OF TWO SUCCESSIVE ADRB'S:

|      |      |                |         |                         |
|------|------|----------------|---------|-------------------------|
| 2051 | 7300 | NOISE, CLA CLL |         |                         |
| 2052 | 1177 | TAD            | ,=100   | /SET TALLY FOR 64 TIMES |
| 2053 | 3020 | DCA            | TEMP0   |                         |
| 2054 | 1022 | TAD            | SW2     | /ENABLE DONE BIT        |
| 2055 | 4521 | ADLM           |         | /LOAD MPX REG           |
| 2056 | 4522 | ADSY           |         | /CONVERT                |
| 2057 | 4524 | ADSK           |         | /DONE FLAG?             |
| 2060 | 5257 | JMP            | ,=1     | /NO                     |
| 2061 | 4523 | ADRB           |         | /YES, READ AD BUFFER    |
| 2062 | 3027 | DCA            | TEMPA   | /STORE                  |
| 2063 | 4523 | ADRB           |         | /RE-READ                |
| 2064 | 3030 | DCA            | TEMPB   | /STORE                  |
| 2065 | 1027 | TAD            | TEMPA   | /COMPARE FOR EQUALITY   |
| 2066 | 7041 | CIA            |         |                         |
| 2067 | 1030 | TAD            | TEMPB   |                         |
| 2070 | 7420 | SNL            |         | /LINK SHOULD BE SET     |
| 2071 | 4776 | JMS            | ERPT2   | /NOT EQUAL              |
| 2072 | 7440 | SZA            |         |                         |
| 2073 | 4776 | JMS            | ERPT2   | /NOT EQUAL              |
| 2074 | 7300 | CLA CLL        |         |                         |
| 2075 | 2026 | ISZ            | TEMP0   | /CONTINUE               |
| 2076 | 5256 | JMP            | NOISE+5 | /YES                    |
| 2077 | 7200 | CLA            |         |                         |
| 2100 | 1041 | TAD            | K207    |                         |
| 01   | 4775 | JMS            | PRLP    | /RING PULL              |
| 02   | 5251 | JMP            | NOISE   | /DO TE AGAIN            |

/ROUTINE TO CHECK FOR NOISE IN MULTIPLE

|      |      |                 |        |
|------|------|-----------------|--------|
| 2103 | 7300 | GLITCH, CLA CLL |        |
| 2104 | 1177 | TAD             | [=100  |
| 2105 | 3026 | DCA             | TEMP0  |
| 2106 | 7604 | LAS             |        |
| 2107 | 0175 | AND             | [17    |
| 2110 | 3031 | DCA             | TEMPC  |
| 2111 | 1031 | TAD             | TEMPC  |
| 2112 | 4521 | ADLM            |        |
| 2113 | 4522 | ADST            |        |
| 2114 | 4524 | ADSK            |        |
| 2115 | 5314 | JMP             | ,=1    |
| 2116 | 4523 | ADRB            |        |
| 2117 | 3027 | DCA             | TEMPA  |
| 2120 | 4344 | CHNL1, JMS      | RANCHN |
| 2121 | 1077 | TAD             | CHNL   |

/OPERATOR TO SELECT CHANNEL

/GET RANDOM CHANNEL

/MAINDEC-08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-20

|      |      |         |        |
|------|------|---------|--------|
| 2122 | 4521 | ADLM    |        |
| 2123 | 4527 | ADRS    |        |
| 2124 | 2026 | ISE     | TEMP0  |
| 2125 | 5320 | JMP     | CHNL1  |
| 2126 | 7300 | CLA CLL |        |
| 2127 | 4523 | ADRB    |        |
| 2130 | 3030 | DCA     | TEMPB  |
| 2131 | 1027 | TAD     | TEMPA  |
| 2132 | 7041 | CIA     |        |
| 2133 | 1030 | TAD     | TEMPB  |
| 2134 | 7420 | SNL     |        |
| 2135 | 4774 | JMS     | ERPT5  |
| 2136 | 7440 | SEA     |        |
| 2137 | 4774 | JMS     | ERPT5  |
| 2140 | 7300 | CLA CLL |        |
| 2141 | 1041 | TAD     | K207   |
| 2142 | 4775 | JMS     | PRLP   |
| 2143 | 5303 | JMP     | GLITCH |

|      |      |             |        |
|------|------|-------------|--------|
| 2144 | 1357 | RANCHN, TAD | FSTNO  |
| 2145 | 7006 | RTL         |        |
| 2146 | 3357 | DCA         | FSTNO  |
| 2147 | 1357 | TAD         | FSTNO  |
| 2150 | 1360 | TAD         | SECNO  |
| 2151 | 7006 | RTL         |        |
| 2152 | 1360 | TAD         | SECNO  |
| 2153 | 7012 | RTR         |        |
| 2154 | 0175 | AND         | [17    |
| 2155 | 3077 | DCA         | CHNL   |
| 2156 | 5744 | JMP I       | RANCHN |

|      |      |        |      |
|------|------|--------|------|
| 2157 | 0437 | FSTNO, | 0437 |
| 2160 | 2525 | SECNO, | 2525 |

/MAINDEC-08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 3

|      |      |
|------|------|
| 2174 | 1756 |
| 2175 | 1534 |
| 2176 | 1710 |

2177 1751  
2200

PAGE

/ROUTINE TO PERFORM 1000(10) CONVERSIONS OF ANY GIVEN VOLTAGE ON SELECTED CHANNEL

```
2200 4466 RESOL, JMS I SETUP
2201 1054 TAD XSTOR
2202 3010 DCA 10
2203 3777 DCA STORAG
2204 4452 JMS I XMOVE /CLEAR WORK AREA
2205 4400 STORAG
2206 4401 STORAG+1
2207 6030 -1750
2210 1165 TAD [-1750
2211 3026 DCA TEMP0
2212 4520 ADCL
2213 7604 LAS /GET CHANNEL
2214 0175 AND [-17
2215 3063 DCA CHAN /STORE CHANNEL
2216 1063 TAD CHAN
2217 4521 ADLM /LOAD CHANNEL
2220 4522 ADST
2221 4524 ADSK
2222 5221 JMP , -1
2223 4523 ADRB
2224 3410 DCA I 10 /PLACE IN TABLE
2225 2026 ISZ TEMP0 /DONE?
2226 5220 JMP , -6 /NO
2227 5455 JMP I XCOMPR /YES, NOW CHECK
2377 4400
2400 PAGE
```

/ROUTINE TO COMPARE FOR GREATER THAN + OR - 1 LSB DIFFERENCE IN 1000(10) CONVERSIONS

```
2400 7300 COMPAR, CLA CLL
2401 1164 TAD [-1747
2402 3026 DCA TEMP0
2403 1054 TAD XSTOR /POINTER FOR FIRST WORD
2404 3010 DCA 10
2405 1410 TAD I 10
2406 3027 DCA TEMPA
2407 7200 COMPR1, CLA
2410 1410 TAD I 10
2411 3030 DCA TEMPB
2412 1027 TAD TEMPA
2413 7041 CIA
```

/MAINDEC=08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 3-1

```
2414 1030 TAD TEMPB
2415 7440 SZA /SKIP HERE
2416 5222 JMP , +4 /AND
2417 7420 SNL /HERE IF =
20 5222 JMP , +2
21 5257 JMP AOK
```

|      |      |       |        |
|------|------|-------|--------|
| 1561 | 1167 | TAD   | [5402  |
| 1562 | 3001 | DCA   | 1      |
| 1563 | 7040 | CMA   |        |
| 1564 | 3062 | DCA   | ERSWIT |
| 1565 | 3767 | DCA I | XIND   |
| 1566 | 5752 | JMP I | XSETUP |
| 1567 | 1346 | XIND, | IND    |

1600 PAGE

/ROUTINE TO CHECK IF TEST COMPLETED ITERATION

|      |      |         |       |                  |
|------|------|---------|-------|------------------|
| 1600 | 0000 | XTAL,   | 0     |                  |
| 1601 | 7604 | LAS     |       |                  |
| 1602 | 0022 | AND     | SW2   | /LOOP OVERRIDE?  |
| 1603 | 7640 | SZA CLA |       |                  |
| 1604 | 5230 | JMP     | XTAL1 | /YES             |
| 1605 | 7604 | LAS     |       |                  |
| 1606 | 0025 | AND     | SW3   | /TEST SELECTED?  |
| 1607 | 7640 | SZA CLA |       |                  |
| 1610 | 5214 | JMP     | ,+4   |                  |
| 1611 | 2034 | ISZ     | TALLY | /DONE WITH TEST? |
| 1612 | 7410 | SKP     |       | /NO              |
| 1613 | 5230 | JMP     | XTAL1 | /YES             |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10 V141

21-MAR-72

13125

PAGE 2-16

|      |      |            |        |                          |
|------|------|------------|--------|--------------------------|
| 1614 | 1062 | TAD        | ERSWIT | /CHECK FOR ERROR         |
| 1615 | 7640 | SZA CLA    |        | /ERROR THIS PASS?        |
| 1616 | 5224 | JMP        | ,+6    | /NO                      |
| 1617 | 1017 | TAD        | MSGPNT | /GET MESSAGE POINTER     |
| 1620 | 1045 | TAD        | M1     | /DECREMENT POINTER       |
| 1621 | 3017 | DCA        | MSGPNT | /RESTORE POINTER         |
| 1622 | 1045 | TAD        | M1     |                          |
| 1623 | 3062 | DCA        | ERSWIT | /RESTORE ERROR INDICATOR |
| 1624 | 1200 | TAD        | XTAL   | /SET RETURN ADDRESS      |
| 1625 | 1046 | TAD        | M2     |                          |
| 1626 | 3200 | DCA        | XTAL   | /STORE RETURN ADDRESS    |
| 1627 | 5600 | JMP I      | XTAL   |                          |
| 1630 | 2017 | XTAL1, ISZ | MSGPNT |                          |
| 1631 | 5600 | JMP I      | XTAL   |                          |

/POINTER FOR SELECTED TEST OPTION

|      |      |       |         |
|------|------|-------|---------|
| 1632 | 0223 | XTST, | TST0-1  |
| 1633 | 0240 |       | TST1-1  |
| 1634 | 0265 |       | TST2-1  |
| 1635 | 0306 |       | TST3-1  |
| 1636 | 0326 |       | TST4-1  |
| 1637 | 0346 |       | TST5-1  |
| 1640 | 0400 |       | TST6-1  |
| 1641 | 0426 |       | TST7-1  |
| 1642 | 0451 |       | TST10-1 |
| 1643 | 0501 |       | TST11-1 |
| 1644 | 0600 |       | TST12-1 |
| 1645 | 0636 |       | TST13-1 |
| 1646 | 0710 |       | TST14-1 |

/ROUTINE TO SELECT SPECIFIC LOGIC TEST SUBROUTINE

|      |      |               |        |                          |
|------|------|---------------|--------|--------------------------|
| 1647 | 0000 | XSELEC, 0     |        |                          |
| 1650 | 7604 | LAS           |        | /GET TEST                |
| 1651 | 0175 | AND           | C17    |                          |
| 1652 | 3026 | DCA           | TEMP0  | /STORE TEST              |
| 1653 | 1026 | TAD           | TEMP0  |                          |
| 1654 | 1045 | TAD           | M1     |                          |
| 1655 | 1166 | TAD           | C140   |                          |
| 1656 | 3017 | DCA           | 17     | /MESSAGE POINTER SET NOW |
| 1657 | 1026 | TAD           | TEMP0  | /GET TEST                |
| 1660 | 1266 | TAD           | JMPLOC | /DEVELOP POINTER         |
| 1661 | 0051 | AND           | K77    |                          |
| 1662 | 1267 | TAD           | JMPINS | /DEVELOP INSTRUCTION     |
| 1663 | 3264 | DCA           | JMPPTR |                          |
| 1664 | 7402 | JMPPTR, HLT   |        | /DO IT!                  |
| 1665 | 7402 | HLT           |        | /TRAP                    |
| 1666 | 1632 | JMPLOC, XTEST |        |                          |
| 1667 | 5600 | JMPINS, 5600  |        |                          |

/ERROR HANDLERS FOR OPEN LOOP TESTS

1670 0000 ERPT1, 0

/MAINDEC=08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 2-17

|      |      |         |         |                    |
|------|------|---------|---------|--------------------|
| 1671 | 7604 | LAS     |         |                    |
| 1672 | 0020 | AND     | SW0     |                    |
| 1673 | 7710 | SPA CLA |         |                    |
| 1674 | 5300 | JMP     | ,+4     |                    |
| 1675 | 4777 | JMS     | MESSAGE |                    |
| 1676 | 4003 | EMSG20  |         |                    |
| 1677 | 4776 | JMS     | CRLF    |                    |
| 1700 | 4775 | JMS     | MESS    |                    |
| 1701 | 4776 | JMS     | CRLF    |                    |
| 1702 | 7604 | LAS     |         |                    |
| 1703 | 0021 | AND     | SW1     | /HALT ON ERROR?    |
| 1704 | 7650 | SNA CLA |         | /SKIP IF YES       |
| 1705 | 5774 | JMP     | RESOL   |                    |
| 1706 | 7402 | HLT     |         |                    |
| 1707 | 5774 | JMP     | RESOL   | /RETURN TO ROUTINE |

|      |      |          |         |                    |
|------|------|----------|---------|--------------------|
| 1710 | 0000 | ERPT2, 0 |         |                    |
| 1711 | 7604 | LAS      |         |                    |
| 1712 | 0020 | AND      | SW0     |                    |
| 1713 | 7710 | SPA CLA  |         |                    |
| 1714 | 5320 | JMP      | ,+4     |                    |
| 1715 | 4777 | JMS      | MESSAGE |                    |
| 1716 | 4034 | EMSG21   |         |                    |
| 1717 | 4776 | JMS      | CRLF    |                    |
| 1720 | 7604 | LAS      |         |                    |
| 1721 | 0021 | AND      | SW1     | /HALT ON ERROR?    |
| 1722 | 7650 | SNA CLA  |         | /SKIP IF YES       |
| 1723 | 5710 | JMP I    | ERPT2   |                    |
| 1724 | 1027 | TAD      | TEMPA   |                    |
| 1725 | 7402 | HLT      |         |                    |
| 1726 | 7200 | CLA      |         |                    |
| 1727 | 1030 | TAD      | TEMPB   |                    |
| 1730 | 7402 | HLT      |         |                    |
| 1731 | 5773 | JMP      | NOISE   | /RETURN TO ROUTINE |

|      |      |         |        |                                       |
|------|------|---------|--------|---------------------------------------|
| 12   | 7430 | SZL     |        |                                       |
| 2423 | 5230 | JMP     | ,+5    |                                       |
| 2424 | 7040 | CMA     |        |                                       |
| 2425 | 7440 | SZA     |        | /SKIP HERE IF DIFFERENCE +1 LSB       |
| 2426 | 7410 | SKP     |        |                                       |
| 2427 | 5257 | JMP     | AOK    |                                       |
| 2430 | 7100 | CLL     |        |                                       |
| 2431 | 7010 | RAR     |        |                                       |
| 2432 | 7440 | SZA     |        | /SKIP HERE                            |
| 2433 | 5237 | JMP     | ,+4    | /AND                                  |
| 2434 | 7420 | SNL     |        | /HERE IF DIFFERENCE -1 LSB            |
| 2435 | 7410 | SKP     |        |                                       |
| 2436 | 5257 | JMP     | AOK    |                                       |
| 2437 | 7300 | CLA CLL |        | /CHECK FOR SPECIAL CASE OF 7777 AND 0 |
| 2440 | 1027 | TAD     | TEMPA  |                                       |
| 2441 | 7440 | SZA     |        | /A=0?                                 |
| 2442 | 7410 | SKP     |        | /NO                                   |
| 2443 | 5247 | JMP     | ,+4    | /YES                                  |
| 2444 | 7040 | CMA     |        | /A=7777?                              |
| 2445 | 7440 | SZA     |        | /SKIP IF YES                          |
| 2446 | 4777 | JMS     | ERPT1  |                                       |
| 2447 | 1030 | TAD     | TEMPB  | /A =7777 OR 0                         |
| 2450 | 7440 | SZA     |        | /B=0?                                 |
| 2451 | 5253 | JMP     | ,+2    | /NO                                   |
| 2452 | 5257 | JMP     | AOK    |                                       |
| 2453 | 7040 | CMA     |        | /B=7777?                              |
| 2454 | 7440 | SZA     |        | /SKIP IF YES                          |
| 2455 | 4777 | JMS     | ERPT1  |                                       |
| 2456 | 5257 | JMP     | AOK    |                                       |
| 2457 | 7300 | CLA CLL |        |                                       |
| 2460 | 1030 | TAD     | TEMPB  |                                       |
| 2461 | 3027 | DCA     | TEMPA  |                                       |
| 2462 | 2026 | ISZ     | TEMP0  | /DONE?                                |
| 2463 | 5207 | JMP     | COMPR1 |                                       |
| 2464 | 2273 | ISZ     | FIVHUN |                                       |
| 2465 | 5776 | JMP     | RESOL  |                                       |
| 2466 | 1375 | TAD     | (-764  | /COUNT OF 500(10).                    |
| 2467 | 3273 | DCA     | FIVHUN |                                       |
| 2470 | 1041 | TAD     | K207   |                                       |
| 2471 | 4774 | JMS     | PRCP   |                                       |
| 2472 | 5776 | JMP     | RESOL  | /YES, REPEAT TEST                     |
| 2473 | 7014 | FIVHUN, | -764   |                                       |
| 2574 | 1534 |         |        |                                       |
| 2575 | 7014 |         |        |                                       |
| 2576 | 2200 |         |        |                                       |
| 2577 | 1670 |         |        |                                       |
|      | 2600 | PAGE    |        |                                       |

/LAB8=E SYSTEM CHECK

|      |      |       |       |
|------|------|-------|-------|
| 2600 | 0000 | SYST, | 0     |
| 2601 | 4466 | JMS I | SETUP |
| 2602 | 4520 | ADCL  |       |
| 2603 | 7402 | HLT   |       |
| 2604 | 7604 | LAS   |       |
| 2605 | 0377 | AND   | (700  |

|      |      |         |         |                                       |
|------|------|---------|---------|---------------------------------------|
| 2606 | 1376 | TAD     | (4040   | /RATE AND ENABLE EXT'L                |
| 2607 | 3031 | DCA     | TEMPC   | /SAVE                                 |
| 2610 | 1031 | TAD     | TEMPC   |                                       |
| 2611 | 4530 | CLOE    |         | /START CLOCK                          |
| 2612 | 7040 | CMA     |         |                                       |
| 2613 | 4532 | CLZE    |         |                                       |
| 2614 | 7200 | CLA     |         |                                       |
| 2615 | 1024 | TAD     | SW4     | /EXT START FOR A/D                    |
| 2616 | 3026 | DCA     | TEMP0   |                                       |
| 2617 | 4775 | JMS     | MESSAGE | /TYPE OUT TEST INSTRUCTIONS           |
| 2620 | 4215 | AUTMSG  |         |                                       |
| 2621 | 7402 | HLT     |         |                                       |
| 2622 | 7604 | LAS     |         |                                       |
| 2623 | 0025 | AND     | SW5     |                                       |
| 2624 | 7440 | SZA     |         | /SKIP IF NOT AUTO-INCREMENT           |
| 2625 | 4321 | JMS     | LS1CHN  | /CHECK FOR LAST CHANNEL               |
| 2626 | 7604 | LAS     |         |                                       |
| 2627 | 0175 | AND     | (17     |                                       |
| 2630 | 4521 | ADLM    |         | /LOAD CHANNEL                         |
| 2631 | 1026 | TAD     | TEMP0   |                                       |
| 2632 | 4526 | ADLE    |         | /LOAD EXT ENABLE BIT IF PRESENT       |
| 2633 | 1026 | TAD     | TEMP0   |                                       |
| 2634 | 7650 | SNA CLA |         | /SKIP FOR EXT'L ENABLE                |
| 2635 | 5245 | JMP     | ,+10    |                                       |
| 2636 | 1374 | TAD     | (7001   | /=X(MAX) TO RESET SWEEP               |
| 2637 | 3027 | DCA     | TEMPA   | /AND START INITIAL CONVERSION         |
| 2640 | 4533 | CLSA    |         | /FROM REAL                            |
| 2641 | 4531 | CLSK    |         | /TIME CLOCK                           |
| 2642 | 5241 | JMP     | ,=1     |                                       |
| 2643 | 7240 | CLA CMA |         |                                       |
| 2644 | 4532 | CLZE    |         | /STOP CLOCK                           |
| 2645 | 7200 | CLA     |         |                                       |
| 2646 | 7410 | SKP     |         |                                       |
| 2647 | 4522 | ADST    |         | /START CONVERSION HERE FOR ALL VALUES |
| 2650 | 4524 | ADSK    |         | /OTHER THAN =X(MAX)                   |
| 2651 | 5250 | JMP     | ,=1     |                                       |
| 2652 | 4527 | ADRS    |         |                                       |
| 2653 | 0175 | AND     | (17     |                                       |
| 2654 | 1030 | TAD     | TEMPB   |                                       |
| 2655 | 7001 | IAC     |         |                                       |
| 2656 | 7440 | SZA     |         |                                       |
| 2657 | 5261 | JMP     | ,+2     |                                       |
| 2660 | 4521 | ADLM    |         |                                       |
| 2661 | 4523 | ADRB    |         | /GET Y VALUE                          |
| 2662 | 4540 | DILY    |         |                                       |
| 2663 | 7200 | CLA     |         |                                       |
| 2664 | 1027 | TAD     | TEMPA   |                                       |

/MAINDEC=08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10 V141

21-MAR-72

13125 PAGE 3-3

|      |      |         |       |                   |
|------|------|---------|-------|-------------------|
| 2665 | 4537 | DILX    |       |                   |
| 2666 | 7001 | IAC     |       | /GET NEXT X VALUE |
| 2667 | 3027 | DCA     | TEMPA |                   |
| 2670 | 1027 | TAD     | TEMPA |                   |
| 2671 | 1374 | TAD     | (7001 |                   |
| 2672 | 7640 | SZA CLA |       | /SKIP IF +X(MAX)  |
| 2673 | 7410 | SKP     |       |                   |
| 2674 | 5305 | JMP     | RESTR |                   |
| 75   | 4536 | DISD    |       |                   |

|      |      |            |        |                              |
|------|------|------------|--------|------------------------------|
| 6    | 5275 | JMP        | ,=1    |                              |
| 2677 | 4541 | DIXY       |        |                              |
| 2700 | 1047 | TAD        | M4     | /TIME OUT TO ALLOW           |
| 2701 | 3340 | DCA        | TEMPX  | /TRACE TO RETURN TO 1001(X)  |
| 2702 | 2340 | ISE        | TEMPX  | /AND SETTLE                  |
| 2703 | 5302 | JMP        | ,=1    |                              |
| 2704 | 5247 | JMP        | STCONV |                              |
| 2705 | 1031 | RESTR, TAD | TEMPC  | /TO RESTART CLOCK            |
| 2706 | 4530 | CLOE       |        |                              |
| 2707 | 7040 | CMA        |        |                              |
| 2710 | 4532 | CLZE       |        |                              |
| 2711 | 7604 | LAS        |        |                              |
| 2712 | 0025 | AND        | SW5    | /A-I MODE                    |
| 2713 | 7640 | SZA CLA    |        | /SKIP IF NO                  |
| 2714 | 5236 | JMP        | CLKST  |                              |
| 2715 | 7604 | LAS        |        |                              |
| 2716 | 0175 | AND        | C17    | /TO CHANGE CHANNEL           |
| 2717 | 4521 | ADLM       |        |                              |
| 2720 | 5236 | JMP        | CLKST  | /GO                          |
| 2721 | 0000 | LSTCHN, 0  |        | /CHECK FOR LAST CHANNEL      |
| 2722 | 7604 | LAS        |        | /IF AUTO INCREMENT MODE      |
| 2723 | 0175 | AND        | C17    |                              |
| 2724 | 7040 | CMA        |        |                              |
| 2725 | 3030 | DCA        | TEMPB  |                              |
| 2726 | 2321 | ISE        | LSTCHN |                              |
| 2727 | 2321 | ISE        | LSTCHN |                              |
| 2730 | 7604 | LAS        |        |                              |
| 2731 | 0025 | AND        | SW5    |                              |
| 2732 | 7650 | SNA CLA    |        | /SKIP IF AUTO INCREMENT MODE |
| 2733 | 5337 | JMP        | ,+4    |                              |
| 2734 | 1024 | TAD        | SW4    |                              |
| 2735 | 1025 | TAD        | SW5    |                              |
| 2736 | 3026 | DCA        | TEMP0  |                              |
| 2737 | 5721 | JMP I      | LSTCHN |                              |
| 2740 | 0000 | TEMPX, 0   |        |                              |
| 2774 | 7001 |            |        |                              |
| 2775 | 1274 |            |        |                              |
| 2776 | 4040 |            |        |                              |
| 2777 | 0700 |            |        |                              |
|      | 3000 | PAGE       |        |                              |
| 3000 | 0000 | MESS, 0    |        |                              |
| 3001 | 4777 | JMS        | CRLF   |                              |
| 3002 | 7300 | CLA CLL    |        |                              |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 3-4

|      |      |         |       |
|------|------|---------|-------|
| 3003 | 1027 | TAD     | TEMPA |
| 3004 | 0376 | AND     | (7000 |
| 3005 | 7002 | BSW     |       |
| 3006 | 7012 | RTR     |       |
| 3007 | 7010 | RAR     |       |
| 3010 | 1375 | TAD     | (260  |
| 3011 | 4774 | JMS     | PRLP  |
| 3012 | 7300 | CLA CLL |       |
| 3013 | 1027 | TAD     | TEMPA |
| 3014 | 7006 | RTL     |       |
| 3015 | 7004 | RAL     |       |
| 3016 | 0376 | AND     | (7000 |

|      |      |         |       |
|------|------|---------|-------|
| 3017 | 7002 | BSW     |       |
| 3020 | 7012 | RTR     |       |
| 3021 | 7010 | RAR     |       |
| 3022 | 1375 | TAD     | (260  |
| 3023 | 4774 | JMS     | PRLP  |
| 3024 | 7200 | CLA     |       |
| 3025 | 1027 | TAD     | TEMPA |
| 3026 | 7012 | RTR     |       |
| 3027 | 7010 | RAR     |       |
| 3030 | 0373 | AND     | (7    |
| 3031 | 1375 | TAD     | (260  |
| 3032 | 4774 | JMS     | PRLP  |
| 3033 | 7300 | CLA CLL |       |
| 3034 | 1027 | TAD     | TEMPA |
| 3035 | 0373 | AND     | (7    |
| 3036 | 1375 | TAD     | (260  |
| 3037 | 4774 | JMS     | PRLP  |
| 3040 | 7300 | CLA CLL |       |
| 3041 | 4777 | JMS     | CRLF  |
| 3042 | 7300 | CLA CLL |       |
| 3043 | 1030 | TAD     | TEMPB |
| 3044 | 0376 | AND     | (7000 |
| 3045 | 7002 | BSW     |       |
| 3046 | 7010 | RAR     |       |
| 3047 | 7012 | RTR     |       |
| 3050 | 1375 | TAD     | (260  |
| 3051 | 4774 | JMS     | PRLP  |
| 3052 | 7300 | CLA CLL |       |
| 3053 | 1030 | TAD     | TEMPB |
| 3054 | 7006 | RTL     |       |
| 3055 | 7004 | RAL     |       |
| 3056 | 0376 | AND     | (7000 |
| 3057 | 7002 | BSW     |       |
| 3060 | 7010 | RAR     |       |
| 3061 | 7012 | RTR     |       |
| 3062 | 1375 | TAD     | (260  |
| 3063 | 4774 | JMS     | PRLP  |
| 3064 | 7300 | CLA CLL |       |
| 3065 | 1030 | TAD     | TEMPB |
| 3066 | 7010 | RAR     |       |
| 3067 | 7012 | RTR     |       |
| 3070 | 0373 | AND     | (7    |
| 3071 | 1375 | TAD     | (260  |

/MAINDEC-08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 3-5

|      |      |         |       |
|------|------|---------|-------|
| 3072 | 4774 | JMS     | PRLP  |
| 3073 | 7300 | CLA CLL |       |
| 3074 | 1030 | TAD     | TEMPB |
| 3075 | 0373 | AND     | (7    |
| 3076 | 1375 | TAD     | (260  |
| 3077 | 4774 | JMS     | PRLP  |
| 3100 | 7300 | CLA CLL |       |
| 3101 | 4777 | JMS     | CRLF  |
| 3102 | 4777 | JMS     | CRLF  |
| 3103 | 7300 | CLA CLL |       |
| 3104 | 5600 | JMP I   | MESS  |

3173 0007

74 1534

15 0260  
3176 7000  
3177 1542  
3200

PAGE

/MAINDEC-28-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4

3200

PAGE

/CONTROL LOGIC ERROR MESSAGES

3200 3736 MSG0, TEXT "++TEST 0 = DONE FLAG OR TIMING ERROR FLAG NOT CLEARED OR SKIP FAILURE++"  
3201 2405  
3202 2324  
3203 4060  
3204 4055  
3205 4004  
3206 1716  
3207 0540  
3210 0614  
3211 0107  
3212 4017  
3213 2240  
3214 2411  
3215 1511  
3216 1607  
3217 4005  
3220 2222  
3221 1722  
3222 4006  
3223 1401  
3224 0740  
3225 1617  
3226 2440  
3227 0314  
3230 0501  
3231 2205  
3232 0440  
3233 1722  
3234 4023  
3235 1311  
3236 2040  
3237 0601  
3240 1114  
3241 2522  
3242 0537  
3243 3600  
3244 3736 MSG1, TEXT "++TEST 1 = DONE FLAG NOT SET THEN CLEARED OR SKIP FAILURE++"  
3245 2405  
3246 2324  
3247 4061  
3250 4055  
3251 4004  
3252 1716  
3253 0540  
3254 0614

3255 0107  
3256 4016  
3257 1724  
3260 4023

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-1

3261 0524  
3262 4024  
3263 1005  
3264 1640  
3265 0314  
3266 0501  
3267 2205  
3270 0440  
3271 1722  
3272 4023  
3273 1311  
3274 2040  
3275 0601  
3276 1114  
3277 2522  
3300 0537  
3301 3600  
3302 3736  
3303 2405  
3304 2324  
3305 4062  
3306 4055  
3307 4024  
3310 1115  
3311 1116  
3312 0740  
3313 0522  
3314 2217  
3315 2240  
3316 0614  
3317 0107  
3320 4016  
3321 1724  
3322 4023  
3323 0524  
3324 4024  
3325 1005  
3326 1640  
3327 0314  
3330 0501  
3331 2205  
3332 0440  
3333 1722  
3334 4023  
3335 1311  
3336 2040  
3337 0601  
3340 1114  
3341 2522  
3342 0537  
3343 3600  
3344 3736  
45 2405

MSG2, TEXT "++TEST 2 = TIMING ERROR FLAG NOT SET THEN CLEARED OR SKIP FAILURE++"

MSG3, TEXT "++TEST 3 = UNEXPECTED INTERRUPT OCCURRED++"

46 2324  
3347 4063

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-2

3350 4055  
3351 4025  
3352 1605  
3353 3020  
3354 0503  
3355 2405  
3356 0440  
3357 1116  
3360 2405  
3361 2222  
3362 2520  
3363 2440  
3364 1703  
3365 0325  
3366 2222  
3367 0504  
3370 3736  
3371 0000  
3372 3736  
3373 2405  
3374 2324  
3375 4064  
3376 4055  
3377 4001  
3400 0422  
3401 0240  
3402 0601  
3403 1114  
3404 0504  
3405 4024  
3406 1740  
3407 1201  
3410 1540  
3411 2422  
3412 0116  
3413 2306  
3414 0522  
3415 4024  
3416 1740  
3417 0103  
3420 3736  
3421 0000  
3422 3736  
3423 2405  
3424 2324  
3425 4065  
3426 4055  
3427 4001  
3430 0422  
3431 2340  
3432 0601  
3433 1114  
3434 0504  
3435 4024  
3436 1740

MSG4, TEXT "++TEST 4 = ADRB FAILED TO JAM TRANSFER TO AC++"

MSG5, TEXT "++TEST 5 = ADRS FAILED TO JAM TRANSFER TO AC++"

3437 1201  
3440 1540  
3441 2422  
3442 0116  
3443 2306  
3444 0522  
3445 4024  
3446 1740  
3447 0103  
3450 3736  
3451 0000  
3452 3736  
3453 2405  
3454 2324  
3455 4066  
3456 4055  
3457 4005  
3460 1601  
3461 0214  
3462 0540  
3463 2205  
3464 0711  
3465 2324  
3466 0522  
3467 4016  
3470 1724  
3471 4020  
3472 2217  
3473 2005  
3474 2214  
3475 3140  
3476 1417  
3477 0104  
3500 0304  
3501 3736  
3502 0000  
3503 3736  
3504 2405  
3505 2324  
3506 4067  
3507 4055  
3510 4006  
3511 0111  
3512 1405  
3513 0440  
3514 2417  
3515 4007  
3516 0516  
3517 0522  
3520 0124  
3521 0540  
3522 1116  
3523 2405  
3524 2222  
3525 2520

MSG6, TEXT "++TEST 6 = ENABLE REGISTER NOT PROPERLY LOADED++"

MSG7, TEXT "++TEST 7 = FAILED TO GENERATE INTERRUPT WITH DONE FLAG++"

3526 2440  
3527 2711  
3530 2410  
3531 4004  
3532 1716  
3533 0540  
3534 0614  
3535 0107  
3536 3736  
3537 0000  
3540 3736  
3541 2405  
3542 2324  
3543 4061  
3544 6040  
3545 5540  
3546 0601  
3547 1114  
3550 0504  
3551 4024  
3552 1740  
3553 0705  
3554 1605  
3555 2201  
3556 2405  
3557 4011  
3560 1624  
3561 0522  
3562 2225  
3563 2024  
3564 4027  
3565 1124  
3566 1040  
3567 2411  
3570 1511  
3571 1607  
3572 4005  
3573 2222  
3574 1722  
3575 4006  
3576 1401  
3577 0737  
3600 3600  
3601 3736  
3602 2405  
3603 2324  
3604 4061  
3605 6140  
3606 5540  
3607 0601  
3610 1114  
3611 0504  
3612 4024  
3613 1740  
3614 1417

EMSG10, TEXT     "-+TEST 10 - FAILED TO GENERATE INTERRUPT WITH TIMING ERROR FLAG+-"

EMSG11, TEXT     "-+TEST 11 - FAILED TO LOAD AND READ MPX REG AND CLEAR AC+-"

3616 4001  
3617 1604  
3620 4022  
3621 0501  
3622 0440  
3623 1520  
3624 3040  
3625 2205  
3626 0740  
3627 0116  
3630 0440  
3631 0314  
3632 0501  
3633 2240  
3634 0103  
3635 3736  
3636 0000  
3637 3736  
3640 2405  
3641 2324  
3642 4061  
3643 6240  
3644 5540  
3645 0601  
3646 1114  
3647 0504  
3650 4024  
3651 1740  
3652 1417  
3653 0104  
3654 4001  
3655 1604  
3656 4022  
3657 0501  
3660 0440  
3661 0114  
3662 1440  
3663 0310  
3664 0116  
3665 1605  
3666 1423  
3667 4011  
3670 1624  
3671 1740  
3672 1520  
3673 3040  
3674 2205  
3675 0737  
3676 3600  
3677 3736  
3700 2405  
3701 2324  
3702 4061  
3703 6340

EMSG12, TEXT     "••TEST 12 - FAILED TO LOAD AND READ ALL CHANNELS INTO MPX REG••"

EMSG13, TEXT     "••TEST 13 - FAILED TO LOAD AND READ ALL CHANNELS IN AUTO-INCREMENT MODE••"

3704 5540  
705 0601  
06 1114

3707 0504  
 3710 4024  
 3711 1740  
 3712 1417  
 3713 0104  
 3714 4001  
 3715 1604  
 3716 4022  
 3717 0501  
 3720 0440  
 3721 0114  
 3722 1440  
 3723 0310  
 3724 0116  
 3725 1605  
 3726 1423  
 3727 4011  
 3730 1640  
 3731 0125  
 3732 2417  
 3733 5511  
 3734 1603  
 3735 2205  
 3736 1505  
 3737 1624  
 3740 4015  
 3741 1704  
 3742 0537  
 3743 3600  
 3744 3736  
 3745 2405  
 3746 2324  
 3747 4061  
 3750 6440  
 3751 5540  
 3752 0601  
 3753 1114  
 3754 0504  
 3755 4024  
 3756 1740  
 3757 0317  
 3760 1520  
 3761 1405  
 3762 2405  
 3763 4003  
 3764 1716  
 3765 2605  
 3766 2223  
 3767 1117  
 3770 1640  
 3771 1116  
 3772 4023

EMSG14, TEXT      "-+TEST 14 - FAILED TO COMPLETE CONVERSION IN SPECIFIED TIME+-"

3773 2005  
 3774 0311  
 3775 0611  
 3776 0504

4000 1115  
 4001 0537  
 4002 3600  
 4003 3736  
 4004 0601  
 4005 1114  
 4006 0504  
 4007 4024  
 4010 1740  
 4011 2205  
 4012 2317  
 4013 1426  
 4014 0540  
 4015 0317  
 4016 1626  
 4017 0522  
 4020 2311  
 4021 1716  
 4022 2340  
 4023 2417  
 4024 4053  
 4025 4017  
 4026 2240  
 4027 5540  
 4030 6140  
 4031 1423  
 4032 0237  
 4033 3600  
 4034 3736  
 4035 2427  
 4036 1740  
 4037 2325  
 4040 0303  
 4041 0523  
 4042 2311  
 4043 2605  
 4044 4022  
 4045 0501  
 4046 0423  
 4047 4016  
 4050 1724  
 4051 4005  
 4052 2125  
 4053 0114  
 4054 3736  
 4055 0000  
 4056 3736  
 4057 0522  
 4060 2217  
 4061 1605

EMSG20, TEXT "++FAILED TO RESOLVE CONVERSIONS TO + OR = 1 LSB++"

EMSG21, TEXT "++TWO SUCCESSIVE READS NOT EQUAL++"

EMSG22, TEXT "++ERRONEOUS EXTERNAL ENABLE OR TIMING ERROR++"

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-8

4062 1725  
 4063 2340  
 4064 0530  
 4065 2405  
 4066 2216  
 4067 0114  
 70 4005

071 1601  
 4072 0214  
 4073 0540  
 4074 1722  
 4075 4024  
 4076 1115  
 4077 1116  
 4100 0740  
 4101 0522  
 4102 2217  
 4103 2237  
 4104 3600  
 4105 3736  
 4106 1517  
 4107 1617  
 4110 2411  
 4111 1611  
 4112 0311  
 4113 2431  
 4114 4006  
 4115 0111  
 4116 1425  
 4117 2205  
 4120 3736  
 4121 0000  
 4122 3736  
 4123 1617  
 4124 1123  
 4125 0540  
 4126 1116  
 4127 4015  
 4130 2514  
 4131 2411  
 4132 2014  
 4133 0530  
 4134 0522  
 4135 4001  
 4136 1604  
 4137 4001  
 4140 5504  
 4141 4002  
 4142 2506  
 4143 0605  
 4144 2237  
 4145 3600

EMSG23, TEXT "MONOTONICITY FAILURE"

EMSG24, TEXT "NOISE IN MULTIPLEXER AND A/D BUFFER"

4146 3736 /END OF LOGIC TEST TYPESTRING  
 4147 0516 XEND, TEXT "END OF LOGIC TEST"

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-9

4150 0440  
 4151 1706  
 4152 4014  
 4153 1707  
 4154 1103  
 4155 4024  
 4156 0523  
 4157 2437  
 4160 3600

4161 3736  
4162 0104  
4163 7005  
4164 4001  
4165 4024  
4166 1740  
4167 0440  
4170 0317  
4171 1626  
4172 0522  
4173 2405  
4174 2234  
4175 4001  
4176 1570  
4177 0540  
4200 1525  
4201 1424  
4202 1120  
4203 1405  
4204 3005  
4205 2240  
4206 0411  
4207 0107  
4210 1617  
4211 2324  
4212 1103  
4213 3736  
4214 0000  
4215 3736  
4216 2305  
4217 2440  
4220 2327  
4221 6540  
4222 5001  
4223 2524  
4224 1735  
4225 1116  
4226 0351  
4227 5440  
4230 4340  
4231 1706  
4232 4003  
4233 1001  
4234 1623  
4235 4011

/HEADER MESSAGE  
XLABEL, TEXT "A-D CONVERTER, AMBE MULTIPLEXER DIAGNOSTIC"

AUTMSG, TEXT "SET SW5 (AUTO-INC), # OF CHANS IN SW8-11, OR SET SW8-11 (SINGLE CHAN)"

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-10

4236 1640  
4237 2327  
4240 7055  
4241 6161  
4242 5440  
4243 1722  
4244 4023  
4245 0524  
4246 4023  
4247 2770  
450 5561

|      |      |
|------|------|
| 0164 | 6031 |
| 0165 | 6030 |
| 0166 | 0140 |
| 0167 | 5402 |
| 0170 | 0076 |
| 0171 | 7760 |
| 0172 | 0475 |
| 0173 | 0445 |
| 0174 | 7761 |
| 0175 | 0017 |
| 0176 | 0322 |
| 0177 | 7700 |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL 10

444

21-MAR-72

13125

PAGE 4-11

[illegible]

|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| 2000 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 2100 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 10000000 | 00001111 |
| 2200 | 11111111 | 11111111 | 11111111 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 2300 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000001 |
| 2400 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11100000 |
| 2500 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00001111 |
| 2600 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 2700 | 11111111 | 11111111 | 11111111 | 11111111 | 10000000 | 00000000 | 00000000 | 00001111 |

|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| 3000 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3100 | 11111000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00011111 |
| 3200 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3300 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3400 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3500 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3600 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 3700 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |

/MAINDEC-08-DHADA=A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10 V141

21-MAR-72

13125

PAGE 4-12

|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| 4000 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 4100 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 4200 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11000000 | 00000000 |
| 4300 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 4400 | 10000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 4500 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |

4600  
4700

5000  
5100

5200  
5300

5400  
5500

5600  
5700

6000  
6100

6200  
6300

6400  
6500  
  
6600  
6700

7000  
7100

7200  
7300

7400  
7500

7600  
7700

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-13

|        |      |        |      |        |      |        |      |
|--------|------|--------|------|--------|------|--------|------|
| ADCL   | 4520 | EMSG6  | 3432 | RESTR  | 2705 | XCLSK  | 0131 |
| ADLE   | 4526 | EMSG7  | 3503 | SCOPE  | 1237 | XCLZE  | 0132 |
| ADLM   | 4521 | EOUT   | 1242 | SECNO  | 2160 | XCOMPR | 0055 |
| ADRB   | 4523 | ERMSG  | 0145 | SELECT | 0065 | XCONVT | 0036 |
| ADRS   | 4527 | ERPT1  | 1670 | SETUP  | 0066 | XDILE  | 0142 |
| ADSE   | 4525 | ERPT2  | 1710 | START  | 0211 | XDILX  | 0137 |
| ADSK   | 4524 | ERPT3  | 1732 | STCONV | 2647 | XDILY  | 0140 |
| ADST   | 4522 | ERPT4  | 1751 | STORAG | 4400 | XDISD  | 0136 |
| AOK    | 2457 | ERPT5  | 1756 | SW0    | 0020 | XDIXY  | 0141 |
| AUTMSG | 4215 | ERR    | 0035 | SW1    | 0021 | XEND   | 4146 |
| AUTO1  | 0643 | ERSWIT | 0062 | SW2    | 0022 | XGLIT  | 0060 |
| AUTO2  | 0674 | ERTYP  | 1226 | SW3    | 0023 | XIND   | 1567 |
| BSW    | 7002 | EXIT   | 1271 | SW4    | 0024 | XINSTR | 0037 |
| CAF    | 6007 | EXT1   | 1032 | SW5    | 0025 | XLABEL | 4161 |
| CHAN   | 0063 | EXTBL  | 0053 | SYST   | 2600 | XMONOT | 0040 |
| CHNL   | 0077 | EXTL   | 1024 | TADDR  | 1224 | XMOVE  | 0052 |
| CHNL1  | 2120 | EXTTE  | 1066 | TAL    | 0064 | XNOISE | 0057 |
| CLAB   | 4535 | FADDR  | 1223 | TALLY  | 0034 | XRESOL | 0056 |
| CLED   | 4534 | FINIS  | 0730 | TEMP0  | 0026 | XSELEC | 1647 |
| CLKST  | 2636 | FIVHUN | 2473 | TEMPA  | 0027 | XSETUP | 1552 |
| CLOE   | 4530 | FSTNO  | 2137 | TEMPB  | 0030 | XSTOR  | 0054 |
| CLSA   | 4533 | GLITCH | 2103 | TEMPC  | 0031 | XSYST  | 0061 |
| CLSK   | 4531 | IND    | 1346 | TEMPO  | 0032 | XTAL   | 1600 |
| CLZE   | 4532 | INITL  | 0216 | TEMPX  | 2740 | XTAL1  | 1630 |
| CNTR1  | 0033 | INSTR  | 1400 | TMG1   | 0475 | XTEST  | 1632 |
| COMPAR | 2400 | JMPINS | 1667 | TST0   | 0224 | XXADCL | 1410 |
| COMPR1 | 2407 | JMPLOC | 1666 | TST1   | 0241 | XXADLE | 1444 |
| CONT   | 2011 | JMPPTR | 1664 | TST10  | 0492 | XXADLM | 1414 |
| CONVT  | 1000 | K1000  | 0050 | TST11  | 0502 | XXADRB | 1424 |
| CRLF   | 1542 | K207   | 0041 | TST12  | 0601 | XXADRS | 1450 |
| DILE   | 4542 | K212   | 0042 | TST13  | 0637 | XXADSE | 1436 |
| DILX   | 4537 | K215   | 0043 | TST14  | 0711 | XXADSK | 1430 |
| DILY   | 4540 | K6500  | 0044 | TST2   | 0266 | XXADST | 1420 |
| DISD   | 4536 | K77    | 0051 | TST3   | 0307 | XXCLAB | 1502 |
| DIXY   | 4541 | LSTCHN | 2721 | TST3S  | 0322 | XXCLED | 1476 |
| DON1   | 0445 | M1     | 0045 | TST4   | 0327 | XXCLOE | 1454 |
| EMSG0  | 3200 | M2     | 0046 | TST5   | 0347 | XXCLSA | 1472 |
| EMSG1  | 3244 | M4     | 0047 | TST6   | 0401 | XXCLSK | 1460 |
| EMSG10 | 3540 | MCTR   | 1225 | TST7   | 0427 | XXCLZE | 1466 |

|        |      |         |      |        |      |        |      |
|--------|------|---------|------|--------|------|--------|------|
| EMSG11 | 3601 | MESSAGE | 1274 | TYPECH | 1314 | XXDILE | 1530 |
| EMSG12 | 3637 | MESS    | 3000 | XADCL  | 0120 | XXDILX | 1514 |
| EMSG13 | 3677 | MONOT   | 2000 | XADLE  | 0126 | XXDILY | 1520 |
| EMSG14 | 3744 | MOVE    | 1200 | XADLM  | 0121 | XXDISD | 1506 |
| EMSG2  | 3302 | MOVEA   | 1213 | XADRB  | 0123 | XXDIXY | 1524 |
| EMSG20 | 4003 | MSGPNT  | 0017 | XADRS  | 0127 |        |      |
| EMSG21 | 4034 | MSRGHT  | 1313 | XADSE  | 0125 |        |      |
| EMSG22 | 4056 | MTP     | 1340 | XADSK  | 0124 |        |      |
| EMSG23 | 4105 | NOISE   | 2051 | XADST  | 0122 |        |      |
| EMSG24 | 4122 | OK      | 2043 | XCLAB  | 0135 |        |      |
| EMSG3  | 3344 | PRLP    | 1534 | XCLED  | 0134 |        |      |
| EMSG4  | 3372 | RANCHN  | 2144 | XCLOE  | 0130 |        |      |
| EMSG5  | 3422 | RESOL   | 2200 | XCLSA  | 0133 |        |      |

/MAINDEC-08-DHADA-A A/D CONVERTER, MULTIPLEXER DIAGNOSTIC

PAL10

V141

21-MAR-72

13125

PAGE 4-14

ERRORS DETECTED: 0

LINKS GENERATED: 53

RUN-TIME: 11 SECONDS

3K CORE USED